Regression-Based Explainable Deep Learning for Estimating Hamiltonian Parameters from Magnetic Nanodot Images

Juan Sebastián Méndez-Rondón[a], María Isabel García-Quimbayo[a], Andrés Marino Álvarez-Meza[a,*], Jorge Iván Montes-Monsalve[b], Jose Darío Agudelo-Giraldo[c]

[a] *AI-Lab, Signal Processing and Recognition Group, Universidad Nacional de Colombia, Sede Manizales, 170003 Manizales, Colombia*
[b] *Dirección Académica, Universidad Nacional de Colombia, Sede de La Paz, 202010 La Paz, Colombia*
[c] *Grupo de Investigación en Física y Matemáticas con Énfasis en la Formación de Ingenieros, Departamento de Física y Matemáticas, Facultad de Ingeniería, Universidad Autónoma de Manizales, Antigua Estación del Ferrocarril, 170002 Manizales, Colombia*

Abstract

Magnetic nanodots host topological spin textures whose stability is dictated by an extended Heisenberg Hamiltonian combining symmetric exchange, the Dzyaloshinskii–Moriya interaction, magnetocrystalline anisotropy, and the Zeeman field. Inferring these atomistic parameters from two-dimensional magnetization images underpins the design of functional magnetic materials, yet it remains an ill-posed inverse problem: under certain external conditions and transient states, configurations relax into noisy, weakly discriminative textures that render the dataset intrinsically ill-conditioned. Deep-learning estimators act as opaque black boxes and rely on slow atomistic simulations. We propose an explainable regression methodology whose novelty lies in data generation and interpretability: a matrix decomposition of the extended Hamiltonian enables GPU-accelerated Metropolis–Hastings Monte Carlo with simulated annealing, accelerating computation without altering the underlying physics, while Regression Activation Maps (RAMs)—a novel error-based attribution metric—turn the network into a traceable gray-box model. Trained on 166,141 configurations, convolutional and Vision Transformer regressors reliably recover the physical parameters, while the higher-order exchange constants remain bounded by the intrinsic identifiability limits of the s_z projection. Comparing both families through their attention maps and accuracy-versus-size trade-off, RAMs isolate the chiral domain walls and skyrmion cores driving predictions.

Keywords: Explainable AI, Magnetic Nanodots, Hamiltonian Parameters, Regression Activation Maps, Deep Learning

1. Introduction

Understanding and controlling magnetic behavior at the nanometric scale is a central challenge in modern condensed matter physics and materials engineering [1]. This capability is particularly relevant to the development of spintronic devices, ultrahigh-density magnetic memories, and neuromorphic computing platforms based on topological magnetic textures [2]. At such length scales, surface effects, geometric confinement, and energy quantization substantially modify magnetic properties relative to the bulk [3]. The resulting enhancement of surface anisotropy, structural imperfections, and thermal fluctuations strongly affects the stability and dynamics of spin configurations [4]. These effects give rise to emergent collective states—including ferromagnetic, helical, labyrinthine, and skyrmionic textures [5]—whose stability is governed by the interplay of competing microscopic interactions [6]. In this context, magnetic nanodots constitute an ideal model platform, as their lateral confinement favors the formation of single-domain states, vortices, and skyrmion-like configurations controlled by the balance among symmetric exchange, magnetic anisotropy, DMI, and external magnetic fields [7]. Therefore, inferring these underlying physical parameters directly from magnetic domain images has become a critical inverse problem for the rational engineering of functional magnetic materials [8].

Within this framework, atomistic Hamiltonian-based modeling constitutes the most rigorous description of the discrete spin interactions, localized dynamics, and boundary effects that dominate nanoscale magnetic systems. Unlike continuum micromagnetic models, which can mask essential surface and finite-size contributions, the extended Heisenberg Hamiltonian explicitly captures the atomistic interactions responsible for stabilizing complex topological textures in magnetic nanodots [9]. In its general form, the Hamiltonian includes several competing terms: the symmetric exchange interaction ($\tilde{J}$ [meV]), which promotes collinear alignment; the DMI ($\tilde{K}_{DM}$ [meV]), which favors chiral and non-collinear spin textures; the magnetocrystalline anisotropy ($\tilde{K}$ [meV]), which selects preferred crystallographic orientations; and the Zeeman coupling to an external magnetic field ($\tilde{H}_{ex}$ [meV]) [10]. The competition among these contributions generates a highly multimodal energy landscape [6], in which physically distinct parameter sets may produce spin states with nearly indistinguishable out-of-plane textures, allowing helical, skyrmion-like, and other metastable

*amalvarezme@unal.edu.co
Email addresses: jumendezro@unal.edu.co (Juan Sebastián Méndez-Rondón), migarciaq@unal.edu.co (María Isabel García-Quimbayo), amalvarezme@unal.edu.co (Andrés Marino Álvarez-Meza), jimontesm@unal.edu.co (Jorge Iván Montes-Monsalve), josed.agudelog@autonoma.edu.co (Jose Darío Agudelo-Giraldo)

configurations to coexist [11]. Compounded by the noise that external conditions and transient states introduce during thermal relaxation, this degeneracy renders the inverse estimation of Hamiltonian parameters from two-dimensional magnetic domain images intrinsically ill-conditioned, imposing a physical limit on parameter identifiability that must be explicitly characterized rather than assumed away [12, 13].

The intrinsic ambiguity of the inverse problem is further exacerbated by the limitations of current experimental and analytical methodologies. Although advanced imaging techniques, including magnetic force microscopy, Lorentz transmission electron microscopy, and spin-polarized scanning tunneling microscopy [14], enable high-resolution visualization of magnetic textures, they typically provide indirect or transformed measurements that are subject to modality-dependent distortions, noise, and finite spatial resolution [15, 16]. In practice, parameter estimation therefore relies heavily on empirical fitting or on comparison with synthetic magnetic configurations generated by atomistic spin simulations or continuum micromagnetic models [17, 18]. Yet similarity measures defined in image space—such as mean squared error or structural similarity—are often inadequate for resolving topologically distinct states, since they primarily quantify visual resemblance rather than physical equivalence. Consequently, the optimization process remains structurally ambiguous, and substantially different Hamiltonian parameter sets may produce magnetic domain patterns that are visually indistinguishable [19, 20].

To circumvent the prohibitive cost of iterative simulation-based optimization, recent efforts have reformulated this inverse problem as a data-driven learning task using deep neural architectures [21]. Although supervised regression models allow fast parameter inference, they generally impose an implicit one-to-one mapping and therefore tend to collapse the intrinsic multimodality of degenerate magnetic systems into over-smoothed point predictions [22]. Methods designed to explicitly capture this uncertainty—such as Bayesian inference, diffusion models, cycle-consistent networks, and physics-informed neural networks—offer improved representational flexibility, but often introduce substantial computational overhead and pronounced optimization stiffness [23, 24, 25]. Beyond these computational challenges, a central limitation remains their opacity. In strongly degenerate parameter spaces, conventional black-box models offer little or no mechanistic understanding of how localized spatial features contribute to the inferred Hamiltonian parameters [26, 27]. Moreover, although gradient-based attribution tools such as class activation maps are widely used in classification, analogous interpretability methods for continuous regression in highly multimodal physical systems remain underdeveloped [28]. This interpretability gap restricts the physical traceability required for robust and scientifically meaningful machine-learning-based inference.

Recent advances have demonstrated the immense potential of deep learning in resolving the inverse problem of magnetic Hamiltonian parameter estimation from domain images. For example, Kwon *et al.* [13] employed ResNet architectures to estimate Dzyaloshinskii-Moriya, anisotropy, and dipolar interactions from Monte Carlo-simulated spin configurations, while Kong *et al.* [8] utilized customized convolutional neural networks to extract exchange stiffness, effective anisotropy, and interfacial DMI from micromagnetic simulations. More recently, Lee *et al.* [21] demonstrated the applicability of fully connected networks for estimating interlayer and intralayer interactions in twisted van der Waals magnets. Despite their impressive predictive accuracy, these pioneering approaches predominantly treat the neural network as a "black box." Specifically, while some works analyze intermediate feature maps [8] or perform statistical error analysis [13, 21], they lack spatially resolved attribution mechanisms to explicitly identify which morphological features of the spin textures drive specific parameter predictions. Furthermore, physical identifiability limits—such as degenerate spin configurations arising from competing symmetric interactions—are often encountered as empirical estimation errors rather than explicitly quantified boundaries. These gaps motivate the explainable, physics-grounded regression methodology developed in this work.

To address these limitations, we propose an inherently explainable framework grounded in an atomistic extended Heisenberg Hamiltonian. The proposed methodology explicitly captures the competition among symmetric exchange, DMI, magnetocrystalline anisotropy, and Zeeman coupling through Metropolis–Hastings Monte Carlo sampling and simulated annealing, enabling the generation of thermodynamically relaxed and highly accurate two-dimensional magnetization fields. These physically grounded data are subsequently integrated with a regression-based deep learning architecture for robust inverse estimation of the governing Hamiltonian parameters. The main contributions of this work are summarized as follows. First, we introduce a matrix/tensor decomposition of the extended Heisenberg Hamiltonian that recasts the atomistic energy as linear-algebra operations, enabling a GPU-accelerated Metropolis–Hastings Monte Carlo with simulated annealing that generates thermodynamically relaxed two-dimensional magnetization images far faster than traditional (Fortran-based) atomistic codes, without altering the underlying physics. Second, we propose Regression Activation Maps (RAMs), a novel error-based attribution method that extends gradient activation mapping to continuous-valued physical targets, turning the regressor into a traceable gray-box model that—together with UMAP/HDBSCAN clustering—links predictions to physically meaningful spin textures such as chiral domain walls and skyrmion cores. Third, we present a comparative study of convolutional and Vision Transformer regressors—contrasting their attention and RAM maps as well as their predictive accuracy relative to model size—and explicitly map the identifiability boundaries of the eight-parameter extended Heisenberg Hamiltonian from the s_z projection, distinguishing reliably recoverable parameters from those that remain degenerate.

To assess the effectiveness of the proposed framework, extensive experiments were performed using a custom-generated dataset of simulated magnetic domain images covering a broad range of Hamiltonian parameter configurations. The obtained results indicate that the proposed method enables accurate inverse prediction of a subset of the Hamiltonian parameters—

most notably the DMI and symmetric exchange—while explicitly exposing the identifiability limits of the remaining terms under the s_z projection, with competitive performance across the evaluated convolutional and transformer models. Moreover, the qualitative and quantitative analyses based on RAMs indicate that the learned representations are associated with physically relevant spin topologies rather than dataset-specific noise or superficial correlations, thereby strengthening the interpretability and reliability of the proposed approach. The remainder of the paper is organized as follows. Section 2 presents the related work. Section 3 details the materials and methods. Section 4 describes the experimental setup. Section 5 presents and discusses the results. Finally, Section 6 concludes the paper and highlights future research directions.

2. Related Work

The inverse estimation of Hamiltonian parameters from magnetic domain configurations is widely recognized as a highly challenging problem due to the complex, nonlinear relationship between physical interactions and observable spin textures. From an experimental standpoint, researchers heavily rely on advanced indirect imaging techniques such as magnetic force microscopy, Lorentz transmission electron microscopy, and spin-polarized scanning tunneling microscopy [14]. While these modalities enable high-resolution visualization of magnetic domains, they inherently capture transformed or partial projections of the underlying three-dimensional spin configurations. Because each technique is governed by distinct physical interaction principles, experimental observations are frequently contaminated by modality-specific distortions, finite spatial resolution limits, and heterogeneous noise, making direct inversion from these measurements generally infeasible [15, 16].

Consequently, practical parameter estimation heavily depends on empirical fitting or direct comparison with synthetic data generated through numerical simulations [29]. Computationally, the problem is most commonly addressed using physics-based forward simulations, specifically atomistic spin models governed by the extended Heisenberg Hamiltonian and continuum micromagnetic models [17, 18, 30]. However, attempting to optimize the inverse mapping by minimizing pixel-level discrepancy functions—such as Mean Squared Error or Mean Absolute Error—frequently fails to capture physically meaningful topological differences or vital spin interactions [31]. Even structural metrics like the Structural Similarity Index fall short in this regard [20]. Ultimately, because the comparison between simulated and experimental data occurs strictly in the image space rather than the parameter space, the optimization landscape becomes highly ambiguous, allowing disparate parameter configurations to yield visually identical domain patterns [19].

To bypass the computationally prohibitive nature of iterative simulation-based optimization, recent advances have increasingly framed the inverse problem as a data-driven mapping utilizing deep learning architectures [21]. The most prevalent approach employs supervised regression models, including convolutional neural networks and transformer-based architectures, to approximate this mapping deterministically. While these models offer rapid inference and scalability, they inherently assume a one-to-one mapping between images and parameters. In highly degenerate magnetic systems, this assumption causes the models to collapse the intrinsic multimodality of the parameter space into a single, often smoothed, point estimate, rendering them highly sensitive to dataset bias and sim-to-real mismatch [22].

To explicitly model uncertainty and capture this multimodality, probabilistic frameworks such as Bayesian inference methods, conditional variational autoencoders, and diffusion models have been introduced [32]. Despite their theoretical flexibility in sampling multiple plausible configurations, these generative and Bayesian approaches suffer from immense computational overhead, unstable training dynamics, and a distinct lack of explicit physical guarantees. In parallel, cycle-consistent frameworks and Physics-Informed Neural Networks attempt to constrain the solution space by enforcing coherence between forward and inverse mappings, or by directly embedding thermodynamic penalties into the loss function [24, 25]. While physically plausible, these hybrid models introduce severe optimization stiffness and rely fundamentally on the accuracy of the underlying forward surrogate, which can dangerously propagate errors within strongly degenerate spin regimes.

Despite the growing predictive capabilities of advanced deep neural networks, their fundamental opacity remains a critical barrier to their widespread adoption in condensed matter physics. In the context of nanoscale magnetism, the delicate energetic competition among the exchange interaction, the DMI, and magnetocrystalline anisotropy creates a densely degenerate parameter space where multiple physical configurations present nearly identical free energies [27]. Traditional black-box deep learning architectures provide no mechanistic insight into how specific spatial features within a magnetic domain image correlate with the predicted Hamiltonian constants [26]. This lack of interpretability prevents researchers from distinguishing whether a model has genuinely captured the underlying physical laws or merely memorized superficial dataset artifacts. While gradient-based attribution methods like Class Activation Maps have been widely adopted for discrete classification tasks, interpreting continuous regression outputs in highly multimodal physical systems remains largely underexplored [28]. The inability to spatially trace and quantify the sources of prediction errors restricts the scientific traceability of these models, underscoring an urgent necessity for architectures that inherently couple high predictive accuracy with transparent, physics-informed spatial interpretations.

To overcome these limitations, our methodology departs from prior work along three concrete axes. First, whereas most existing approaches operate at the continuum micromagnetic scale, we adopt an atomistic description at the nanometric scale, grounded in a rigorous extended Heisenberg Hamiltonian that explicitly resolves the discrete spin interactions and the surface and finite-size effects governing magnetic nanodots. Second, in contrast to the predominantly black-box models used in this field—which expose no mechanism to trace how spatial features drive the inferred parameters—we introduce RAMs, an error-based attribution method that correlates localized spatial activations with the absolute prediction error of the continuous physical targets, turning the regressor into a transparent, physically traceable gray-box model. Third, rather than relying on conventional classical simulation software, we build the data-generation process from a matrix/tensor (algebraic) decomposition of the extended Hamiltonian that, coupled with GPU computing, substantially accelerates the Metropolis–Hastings Monte Carlo with simulated annealing without altering the underlying physics. Together, these choices couple physics-grounded, high-throughput simulation with an inherently explainable regression model, enabling the precise isolation of the degenerate magnetic textures—such as chiral domain walls and skyrmions—that drive specific parameter estimations, and establishing a reliable, transparent, and traceable pathway for analyzing complex nanoscale magnetic systems. Table 1 contrasts these distinctions with representative deep-learning approaches for magnetic Hamiltonian parameter estimation across six technical dimensions.

Table 1: Comparative analysis of deep learning approaches for magnetic Hamiltonian parameter estimation. The works of Kong *et al.* [8], Kwon *et al.* [13], and Lee *et al.* [21] are contrasted against the proposed framework across six technical dimensions.

Dimension	**Kong [8]**	**Kwon [13]**	**Lee [21]**	**Ours**
Hamiltonian terms	Exchange A, aniso. K_{eff}, DMI D	Exchange J, DMI β, dipolar D, aniso. K	Intralayer J, single-ion A, interlayer $J^{\perp}$	Ext. Heisenberg $\tilde{J}_1$–$\tilde{J}_4$, $\tilde{K}_{DM}$, $\tilde{H}_{ex}$, $\tilde{K}_{an1}$, $\tilde{K}_{anS}$
Simulation	Micromagnetic (mumax3, LLG)	MC simulated annealing	Atomistic (stochastic LLG)	Atomistic MC (Metropolis + annealing, multi-GPU)
DL architecture	Custom CNN	ResNet18/50, CNN	MLPs	CNN baseline + DenseNet, ViT, Xception, ResNet
Params. estimated	4	3	3	8
Metrics	MAE, R^2	MAE, PCA of errors	MAPE, MSE	R^2, MAE + UMAP/HDBSCAN clustering
Interpret.	Feature maps (black box)	Black box	Black box (PCA)	**RAMs** (layer/ structure) + identifiability

3. Materials and Methods

3.1. Atomistic Hamiltonian Framework

The theoretical foundation for modeling interacting magnetic dipole moments originates from the seminal formulations of Ernst Ising and Werner Heisenberg [33]. Ising introduced a one-dimensional statistical mechanics model wherein localized spins were heavily constrained to discrete, scalar states (i.e., rigidly pointing "up" or "down"). While the Ising model provided early fundamental insights into macroscopic ferromagnetism and phase transitions, its strict uni-axial constraint renders it structurally insufficient for capturing the continuous rotational degrees of freedom required to model complex, multi-dimensional magnetic textures. This limitation was subsequently resolved by Heisenberg, who formulated a generalized model allowing spin vectors to vary continuously in three-dimensional space with rotational symmetry. While the foundational Heisenberg model successfully captures isotropic exchange driven by the Pauli exclusion principle, modern nanoscale applications—such as the study of magnetic vortices and topologically protected skyrmions in confined nanodot geometries—necessitate an extended approach [2]. This extended framework rigorously incorporates symmetry-breaking relativistic spin-orbit coupling effects, primarily the DMI and magnetocrystalline anisotropy, alongside standard Zeeman coupling [34].

Formally, let $\mathcal{V} = \{1, \ldots, N\}$ denote the set of indices for the discrete magnetic lattice sites. The thermodynamic state of the system is characterized by the spatial distribution of localized magnetic moments, represented by the normalized spin vectors $\mathbf{s}_i \in \mathbb{R}^3$, subject to the constraint $\|\mathbf{s}_i\|_2 = 1$, for $i \in \mathcal{V}$. In a generalized theoretical context, all pairwise symmetric and antisymmetric interactions, as well as site-specific field and anisotropy alignments, can be collapsed into a unified geometric approach [35, 36]. The total effective energy of the system $\mathcal{H} \in \mathbb{R}$ [meV] is thus defined by the following generalized quadratic form:

$$\begin{aligned}\mathcal{H}\left(\mathbf{S}|\breve{\theta}\right) = &-\sum_{i=1}^{N}\sum_{j\in\mathcal{N}(i)}\langle \mathbf{s}_i, \tilde{\mathbf{M}}_{ij}\mathbf{s}_j\rangle \\ &-\sum_{i=1}^{N}\langle \tilde{\mathbf{h}}_i, \mathbf{s}_i\rangle - \sum_{i=1}^{N}\langle \mathbf{s}_i\tilde{\mathbf{K}}_i, \mathbf{s}_i\rangle,\end{aligned} \tag{1}$$

where $\mathcal{N}(i)$ defines the set of interacting nearest neighbors for a given site i, and the matrix $\mathbf{S} \in \mathbb{R}^{N\times 3}$ gathers the continuous spin vectors across the lattice with $\mathbf{s}_i \in \mathbf{S}$. Here, $\breve{\theta} = \{\tilde{\mathbf{M}}_{ij}, \tilde{\mathbf{h}}_i, \tilde{\mathbf{K}}_i\}$, where $\tilde{\mathbf{M}}_{ij} \in \mathbb{R}^{3\times 3}$ represents the complete bilinear interaction matrix coupling neighboring spins, $\tilde{\mathbf{h}}_i \in \mathbb{R}^3$ accounts for linear external field interactions, and $\tilde{\mathbf{K}}_i \in \mathbb{R}^{3\times 3}$ is the anisotropy tensor governing site-specific directional preferences. Casting every interaction as dense linear-algebra operations acting on the spin matrix $\mathbf{S}$ is a deliberate modeling choice: it recasts the atomistic energy as batched matrix and tensor products rather than the sequential site-by-site loops of conventional atomistic codes, which is precisely what enables the GPU-accelerated evaluation and sampling detailed in Section 3.2, without altering the underlying physics.

Afterward, to recover the isolated physical phenomena and define an extended target scalar parameter space $\boldsymbol{\theta} = [\tilde{J}_1, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{DM}, \tilde{H}_{ex}, \tilde{K}_{an1}, \tilde{K}_{anS}]$, the generalized matrices must be mathematically decomposed. By factorizing these parameters, we can map each matrix operation to its specific energetic contribution:

– *Bilinear Coupling ($\tilde{\mathbf{M}}_{ij}$):* A real square matrix can be separated into its symmetric and skew-symmetric components. Here, $\tilde{\mathbf{M}}_{ij}$ is parameterized by factorizing the isotropic exchange scalar $\tilde{J}_r$ for the r-th neighbor shell and the DMI magnitude $\tilde{K}_{DM}$ along a predefined spatial direction vector $\hat{\mathbf{d}}_{ij} \in \mathbb{R}^3$ ($\|\hat{\mathbf{d}}_{ij}\|_2 = 1$), yielding:

$$\tilde{\mathbf{M}}_{ij}(\tilde{J}_r, \tilde{K}_{DM}) = \tilde{J}_r \mathbf{I} - \tilde{K}_{DM}[\hat{\mathbf{d}}_{ij}]_\times, \tag{2}$$

where $\mathbf{I} \in \mathbb{R}^{3\times3}$ represents the identity matrix and $[\hat{\mathbf{d}}_{ij}]_\times \in \mathbb{R}^{3\times3}$ denotes the skew-symmetric matrix representation of the cross-product. The symmetric exchange is modeled across four neighbor shells $r \in \{1, 2, 3, 4\}$, where $\mathcal{N}_r(i)$ denotes the set of neighbors at shell r around site i, ordered by increasing interatomic distance ($d_1 < d_2 < d_3 < d_4$). To resolve the fundamental scale ambiguity of the inverse problem—whereby multiplying all parameters by a common factor leaves the observable spin configuration unchanged—the first-shell exchange is fixed as the energy reference unit, $\tilde{J}_1 \in \mathbb{R}$ in meV, while $\tilde{J}_2, \tilde{J}_3, \tilde{J}_4 \in \mathbb{R}$ are free parameters (also in meV). The DMI acts exclusively between first-shell neighbors and is therefore indexed only over $\mathcal{N}_1(i)$.

– *Zeeman Interaction ($\tilde{\mathbf{h}}_i$):* The linear field term is factorized by extracting the scalar field magnitude $\tilde{H}_{ex}$ and a fixed normalized direction vector $\hat{\mathbf{h}}$, with $\|\hat{\mathbf{h}}\|_2 = 1$, of the externally applied magnetic field, yielding $\tilde{\mathbf{h}}_i = \tilde{H}_{ex}\,\hat{\mathbf{h}} \in \mathbb{R}^3$. The orientation $\hat{\mathbf{h}}$ is held constant within each simulation but is selected among the principal crystallographic axes—either in-plane ($\hat{\mathbf{e}}_x$) or out-of-plane ($\hat{\mathbf{e}}_z$)—to stabilize distinct families of topological textures. Fixing this direction eliminates the orientational degree of freedom of $\tilde{\mathbf{h}}_i$, ensuring that $\tilde{H}_{ex}$ alone serves as the free parameter governing the Zeeman contribution.

– *Magnetocrystalline Anisotropy ($\tilde{\mathbf{K}}_i$):* Modeling cubic magnetocrystalline anisotropy requires factorizing a site-dependent scalar constant $\tilde{K}_i^{eff}$ to scale a quartic interaction. The principal crystallographic axes of the nanodot are defined by the orthonormal basis vectors $\mathbf{e}_{\hat{x}}, \mathbf{e}_{\hat{y}}, \mathbf{e}_{\hat{z}} \in \mathbf{E}$, with $\mathbf{E} \in \mathbb{R}^{3\times3}$, and the directional cosines are given by $s_{im} = \langle \mathbf{e}_m, \mathbf{s}_i \rangle$ ($\forall m \in \{\hat{x}, \hat{y}, \hat{z}\}$). Because atoms at the nanodot surface experience broken inversion symmetry—having fewer neighbors on the exterior side—their anisotropy differs fundamentally from that of interior atoms. The lattice is therefore partitioned into two disjoint subsets: the bulk region $\mathcal{V}_{bulk} \subset \mathcal{V}$, comprising interior sites with complete coordination shells, and the surface region $\mathcal{V}_{surf} \subset \mathcal{V}$, comprising boundary sites exposed to the geometric confinement, with $\mathcal{V}_{bulk} \cup \mathcal{V}_{surf} = \mathcal{V}$ and $\mathcal{V}_{bulk} \cap \mathcal{V}_{surf} = \emptyset$. This yields the site-dependent effective anisotropy constant:

$$\tilde{K}_i^{eff} = \begin{cases} \tilde{K}_{an1} & \text{if } i \in \mathcal{V}_{bulk}, \\ \tilde{K}_{anS} & \text{if } i \in \mathcal{V}_{surf}, \end{cases} \tag{3}$$

where $\tilde{K}_{an1}, \tilde{K}_{anS} \in \mathbb{R}$ are scalar constants measured in meV per atom [37].

Substituting these factorized relationships back into Equation 1 yields the following explicit formulation for the total extended Hamiltonian:

$$\begin{aligned} \mathcal{H}\left(\mathbf{S}|\hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}; \boldsymbol{\theta}\right) = & -\sum_{r=1}^{4} \tilde{J}_r \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_r(i)} \langle \mathbf{s}_i, \mathbf{s}_j \rangle \\ & + \tilde{K}_{DM} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_1(i)} \langle \hat{\mathbf{d}}_{ij}, (\mathbf{s}_i \times \mathbf{s}_j) \rangle \\ & - \tilde{H}_{ex} \sum_{i=1}^{N} \langle \hat{\mathbf{h}}, \mathbf{s}_i \rangle \\ & + \frac{1}{S_o^4} \sum_{i=1}^{N} \tilde{K}_i^{eff} \left(s_{i\hat{x}}^2 s_{i\hat{y}}^2 + s_{i\hat{x}}^2 s_{i\hat{z}}^2 + s_{i\hat{y}}^2 s_{i\hat{z}}^2 \right), \end{aligned} \tag{4}$$

with $\tilde{J}_1 = 1.0$ meV fixed as the energy reference unit. The cubic anisotropy form in the last term follows directly from expanding $\frac{1}{2}(1 - \|\mathbf{s}_i\mathbf{E}\|_4^4)$ under the L_2 normalization constraint $\|\mathbf{s}_i\|_2 = 1$, which yields $s_{i\hat{x}}^2 s_{i\hat{y}}^2 + s_{i\hat{y}}^2 s_{i\hat{z}}^2 + s_{i\hat{z}}^2 s_{i\hat{x}}^2$, normalized by S_o^4 to maintain dimensional consistency across spin magnitudes.

Physically, the extended parameter space $\boldsymbol{\theta}$ actively governs the competing energetic contributions that shape the magnetic energy landscape. Each $\tilde{J}_r$ dictates the collinear alignment tendency between spin pairs separated by the corresponding distance [38]: positive values ($\tilde{J}_r > 0$) favor parallel, ferromagnetic alignment, while negative values ($\tilde{J}_r < 0$) drive antiparallel, antiferromagnetic order. This multi-shell formulation is essential in confined nanodot geometries, where longer-range exchange pathways–mediated through intermediate atomic orbitals–contribute non-negligibly to the global energetic competition and can selectively stabilize or destabilize complex topological textures beyond what first-shell interactions alone would predict [36]. Among these constants, $\tilde{J}_1$ plays a privileged role: it corresponds to the strongest and most spatially direct exchange pathway, coupling each spin to its nearest neighbors in the cubic lattice and establishing the dominant energy scale of the magnetic system. For this reason, $\tilde{J}_1$ is fixed to 1.0 meV throughout all simulations, serving as the absolute energy reference unit against which all remaining interaction strengths are expressed. The remaining constants $\tilde{J}_2$, $\tilde{J}_3$, and $\tilde{J}_4$ are free parameters whose magnitudes decay with increasing shell distance, consistent with the exponential attenuation of quantum mechanical exchange integrals with interatomic separation [38].

The DMI magnitude, represented by $\tilde{K}_{DM}$, is responsible for driving chiral, non-collinear spin winding in systems exhibiting broken spatial inversion symmetry [39]. By actively competing with the dominant first-shell exchange $\tilde{J}_1$, this antisymmetric contribution stabilizes complex topological textures, such as chiral domain walls and skyrmions. It acts exclusively between first-shell neighbors, as longer-range antisymmetric exchange pathways are negligibly small compared to the dominant spin-orbit coupling at nearest-neighbor interfaces. Measured in meV per interacting spin pair, its magnitude is fundamentally dependent on the strength of the spin-orbit coupling at the material's interfaces and is naturally expressed as a fraction of the first-shell exchange energy $\tilde{J}_1$ [39]; the specific sampled range and its physical justification are detailed in Section 4.

The Zeeman coupling energy, parameterized as $\tilde{H}_{ex}$, quantifies the interaction between the localized atomic spins and the externally applied magnetic field [9]. It systematically penalizes magnetic misalignment with the fixed field axis $\hat{\mathbf{h}}$, frequently serving as the primary thermodynamic driving force for macroscopic magnetic phase transitions. This parameter is expressed in meV per atom, mathematically representing the effective energy equivalence ($g\mu_B B$) that transitions a macroscopic magnetic field B measured in Tesla into the atomistic energy regime; with $g \approx 2$ this corresponds to ≈ 0.116 meV per Tesla, so that the sampled range maps directly onto laboratory-accessible fields (Section 4). Also, fixing the field direction along $\hat{\mathbf{h}}$ reduces the dimensionality of the inverse problem by eliminating the orientational degree of freedom of $\tilde{\mathbf{h}}_i$, ensuring that $\tilde{H}_{ex}$ alone governs the Zeeman contribution as a single identifiable scalar. Notably, when the field is applied out-of-plane ($\hat{\mathbf{h}} = \hat{\mathbf{e}}_z$), the Zeeman term couples directly to the observable s_z component, which—as discussed below—renders $\tilde{H}_{ex}$ recoverable from the two-dimensional projection.

Finally, the magnetocrystalline anisotropy encodes the intrinsic directional preference for magnetization relative to the underlying crystal lattice [37]. Arising from relativistic spin-orbit coupling, this term rigorously establishes the energetically favorable (easy) and unfavorable (hard) crystallographic axes for spin orientation within the nanodot. Both constants are measured in meV per atom and represent a relatively subtle symmetry-breaking energy compared to the dominant isotropic exchange. The bulk anisotropy constant $\tilde{K}_{an1}$ governs interior lattice sites $i \in \mathcal{V}_{bulk}$, where each atom is fully coordinated by neighbors in all spatial directions, yielding a locally symmetric energetic environment. Its magnitude ranges from near-zero in soft magnets to several meV per atom in strongly anisotropic interfacial systems [37, 56], with the sampled range specified in Section 4. The surface anisotropy constant $\tilde{K}_{anS}$, by contrast, governs boundary sites $i \in \mathcal{V}_{surf}$, where broken spatial inversion symmetry—arising from the absence of neighbors on the exterior side of the nanodot—produces an asymmetric local environment that intensifies the directional preference for spin orientation. Consequently, $|\tilde{K}_{anS}|$ is generally larger than $|\tilde{K}_{an1}|$, reflecting the enhanced symmetry-breaking at the nanodot boundary. This two-constant formulation is essential for accurately capturing the finite-size and surface-dominated effects that critically govern the stability of emergent topological textures in confined nanoscale geometries [2, 34].

Although the parameter vector $\boldsymbol{\theta}$ governs the full three-dimensional spin field, the observable exploited in this work is restricted to its out-of-plane projection, the s_z component. This choice is dictated by experimental measurability: the most widely available magnetic imaging modalities—such as magnetic force microscopy and the polar magneto-optical Kerr effect—are primarily sensitive to the out-of-plane magnetization [51], whereas resolving the in-plane components requires substantially more specialized techniques. Working on s_z therefore keeps the framework consistent with the information realistically accessible from experiment. This restriction, however, imposes a theoretical limit on parameter identifiability: distinct parameter sets—most notably those related by the sign of the antisymmetric exchange, which reverses the in-plane chirality while leaving s_z invariant—become indistinguishable in the observation space, so that a regressor exposed to both would collapse its predictions toward a degenerate average. To avoid this pathology, the present study deliberately samples the Hamiltonian parameters over value ranges that remain well-posed under the s_z-only observation, ensuring that the retained configurations do not induce the aforementioned averaging saturation; the specific bounds and the associated sign convention are detailed in Section 4.

3.2. Image-Based Dataset Generation From Monte Carlo Simulations

To accurately capture the physical thermodynamic behavior and algorithmically drive the system toward a low-energy, thermodynamically stable configuration of the nanodot, thermal fluctuations must be formally introduced through the framework of statistical mechanics. While the atomistic Hamiltonian $\mathcal{H}$ rigorously defines the internal magnetic energy for a given spatial configuration of spins, it evaluates the system strictly as a frozen, zero-temperature snapshot. Hence, temperature is fundamentally a macroscopic parameter that governs the statistical distribution of the discrete magnetic microstates rather than acting as a localized interaction within the Hamiltonian. Consequently, temperature T $[K]$ couples to the system via the Boltzmann distribution [36]. Formally, a magnetic microstate μ represents a specific, instantaneous realization of the continuous spin matrix, denoted as $\mathbf{S}_\mu$. The total thermodynamic energy of this specific configuration is evaluated against the static environmental vectors and physical interaction strengths, such that $\mathcal{H}(\mu) \equiv \mathcal{H}\left(\mathbf{S}_\mu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta}\right)$. At thermal equilibrium, the probability $P(\mu)$ of the system occupying this specific microstate is defined as:

$$P(\mu) = \frac{1}{\mathcal{Z}} \exp\left(-\frac{\mathcal{H}(\mu)}{k_B T}\right), \tag{5}$$

where k_B is the Boltzmann constant (expressed in meV/K). The term $\mathcal{Z}$ represents the canonical partition function, which mathematically enforces probability normalization by summing over all valid spin configurations ν (i.e., all possible state matrices $\mathbf{S}_\nu$ strictly satisfying the geometric boundary constraint $\|\mathbf{s}_i\|_2 = 1$):

$$\mathcal{Z} = \sum_{\mathbf{S}_\nu} \exp\left(-\frac{\mathcal{H}\left(\mathbf{S}_\nu | \hat{\mathbf{d}}_{ij}, \hat{\mathbf{h}}, \mathbf{E}; \boldsymbol{\theta}\right)}{k_B T}\right). \tag{6}$$

In our computational framework, this thermodynamic distribution is actively sampled using the Metropolis-Hastings Monte Carlo algorithm combined with a simulated annealing protocol [40]. This stochastic sampling ensures a rigorous exploration of the highly multimodal energy landscape characteristic of these degenerate magnetic systems. When a localized spin perturbation proposes a physical transition from the current microstate $\mu^{(t)}$ (characterized by $\mathbf{S}_{\mu^{(t)}}$) to a newly modified state $\mu^{(t+1)}$ (characterized by $\mathbf{S}_{\mu^{(t+1)}}$), the resulting change in the system's total energy is computed as the scalar difference $\Delta\mathcal{H} = \mathcal{H}(\mu^{(t+1)}) - \mathcal{H}(\mu^{(t)})$. To optimize computational efficiency, $\Delta\mathcal{H}$ is evaluated locally at the perturbed lattice site. The proposed state is subsequently accepted or rejected based on the precise transition probability:

$$P(\mu^{(t)} \to \mu^{(t+1)}) = \begin{cases} 1 & \text{if } \Delta\mathcal{H} < 0 \\ \exp\left(-\frac{\Delta\mathcal{H}}{k_B T^{(t)}}\right) & \text{if } \Delta\mathcal{H} \geq 0. \end{cases} \tag{7}$$

Physically, the instantaneous temperature parameter $T^{(t)}$ actively scales the energetic barriers within the system's vast topological energy landscape during the annealing process [40]. At elevated temperatures, the available thermal energy $k_B T^{(t)}$ frequently exceeds local energy penalties ($\Delta\mathcal{H}$), allowing the simulation to accept thermodynamically unfavorable spin alignments. This thermal noise provides the critical kinetic mechanism required to break out of metastable, localized energy minima, such as misaligned ferromagnetic domains [41]. As $T^{(t)}$ is systematically scaled down during the annealing procedure, the probability of accepting higher-energy configurations undergoes exponential decay. This controlled thermal relaxation drives the continuous spin vector field toward a low-energy, thermodynamically stable configuration consistent with the annealing schedule, promoting the stabilization of complex chiral skyrmion textures driven by the underlying DMI. We note that while simulated annealing with a finite cooling rate does not formally guarantee convergence to the absolute global minimum—particularly in DMI-driven systems where numerous metastable chiral states may coexist—the monotonic energy decrease observed across all simulated configurations (see Section 5) provides empirical evidence of thorough thermodynamic relaxation within the sampled parameter space.

Moreover, to ensure that each simulation consistently originates from a fully randomized paramagnetic phase regardless of the specific parameter configuration, the initial annealing temperature (i.e., $T^{(t)}$ at $t = 0$) is established as an *exchange-normalized initial temperature*. Consistent with foundational statistical mechanics [36], this starting temperature is dynamically rescaled in direct proportion to the dominant symmetric exchange energy ($\tilde{J}$). This energetic normalization guarantees that the initial thermal fluctuations are strictly sufficient to overcome the highest localized energetic barriers ($\Delta\mathcal{H}$) for that specific Hamiltonian. This effectively erases initial state bias and prevents the continuous spin vector field from prematurely collapsing into metastable local minima before the controlled cooling protocol begins.

The stochastic sampling described above is realized as a fully vectorized, GPU-resident computation, which constitutes one of the core methodological contributions of this work. Leveraging the matrix formulation of Section 3.1, the local energy variation $\Delta\mathcal{H}$ is not evaluated site by site—as in conventional atomistic simulators—but for the entire spin lattice at once: the neighbor contributions are gathered through tensor-shift operations and combined as batched matrix and tensor products in a single pass. To update many spins simultaneously while preserving the validity of the Markov chain, the lattice is split into two interpenetrating sublattices via a checkerboard decomposition ($(x+y+z) \bmod 2$), so that all mutually non-adjacent sites of a given sublattice are proposed and accepted in parallel within each Metropolis substep. To obtain reliable thermodynamic averages, a large ensemble of independent stochastic replicas is propagated concurrently as an additional batch dimension, and the complete annealing trajectory—both thermalization and measurement sweeps—is just-in-time compiled and executed entirely on-device, eliminating host–device transfers at every sweep. Finally, the independent parameter configurations are distributed across multiple GPUs following a single-program multiple-data scheme, so that entire families of nanodot simulations are relaxed simultaneously. This matrix-based, massively parallel formulation reduces the wall-clock cost of dataset generation by orders of magnitude relative to sequential atomistic codes, without introducing any approximation to the underlying physics.

To generate a robust and diverse dataset for deep learning evaluation, the magnetic parameters governing the Hamiltonian were systematically varied across the simulations. Key variables included the exchange interaction constant ($\tilde{J}$), the magnetocrystalline anisotropy ($\tilde{K}$), the external Zeeman field ($\tilde{H}_{ex}$), and the DMI strength ($\tilde{K}_{DM}$), alongside variations in the base initial temperature ($T^{(0)}$). Subsequently, the transformation of these raw, converged 3D spin states into normalized 2D image arrays followed a strict procedural pipeline:

– *Geometry parsing:* Spatial coordinates defining the nanodot geometry (with a fixed disk radius and thickness as Magnetic Unit Cell (MUC) measure) were established using a binary circular lattice mask.

– *Spin state mapping:* The topological structure was projected into two dimensions by extracting the out-of-plane s_z component of the continuous spin vectors ($\mathbf{s}_i$) across the valid lattice coordinates.

– *Image rendering:* The isolated configurations were rasterized onto a fixed $\breve{H} \times \breve{W}$ pixel grid. A divergent red–blue colormap was utilized to strictly encode opposite spin polarities, visually capturing the chiral winding from the nanodot's core to its perimeter.

– *Preprocessing:* Pixel intensities corresponding to the s_z magnitude were normalized to a strict interval $[0, 1]$ to ensure stable gradient flows during the deep learning optimization phase. Since every component of the target vector $\boldsymbol{\theta}$ is a scalar magnitude that remains invariant under the discrete rotational symmetries of the lattice, the dataset is stored in its raw form and augmented on the fly at training time through random 90° rotations (Section 3.3); being proper rotations, these operations enlarge the effective sample diversity while strictly preserving both the physical parameters and the chiral handedness dictated by the DMI.

Lastly, the simulation pipeline yields a comprehensive image-based dataset, formalized as $\mathcal{D} = \{\mathbf{X}_n, \boldsymbol{\theta}_n\}_{n=1}^{\breve{N}}$, which serves as the final collection of input-output pairs for the deep learning parameter estimation task. Each input tensor

$\mathbf{X}_n \in [0,1]^{\breve{H}\times\breve{W}}$ encodes the two-dimensional spatial magnetization field (the normalized s_z projection), while the corresponding target vector $\boldsymbol{\theta}_n \in \mathbb{R}^{\breve{P}}$ stores the $\breve{P}$-dimensional array of generating thermodynamic and Hamiltonian parameters (e.g., $\boldsymbol{\theta}_n = [T_n^{(0)}, \tilde{J}_{1,n}, \tilde{J}_{2,n}, \tilde{J}_{3,n}, \tilde{J}_{4,n}, \tilde{K}_{DM,n}, \tilde{H}_{ex,n}, \tilde{K}_{an1,n}, \tilde{K}_{anS,n}]$). Figure 1 depicts the main flowchart of our image-based magnetic dataset generation approach.

3.3. Deep Regression Framework for Hamiltonian Parameter Estimation

The inverse estimation of the Hamiltonian parameters and thermodynamic state from magnetic domain images is formalized as a continuous mapping problem. Given the generated dataset $\mathcal{D}$, where $\mathbf{X}_n$ represents the normalized s_z spatial magnetization field and $\boldsymbol{\theta}_n$ represents the vector of generating physical parameters, we aim to learn a parameterized non-linear mapping function $f_\Theta : \mathbb{R}^{\breve{H}\times\breve{W}} \to \mathbb{R}^{\breve{P}}$, to find the predicted parameters $\hat{\boldsymbol{\theta}}_n \in \mathbb{R}^{\breve{P}}$.

Here, f_Θ denotes a deep neural network architecture parameterized by a set of learnable weights and biases Θ. The optimal parameter set $\hat{\Theta}$ is obtained by minimizing a continuous loss function over the training distribution [42]:

$$\hat{\Theta} = \arg\min_{\Theta} \frac{1}{\breve{N}} \sum_{n=1}^{\breve{N}} \mathcal{L}(\boldsymbol{\theta}_n, f_\Theta(\mathbf{X}_n)), \tag{8}$$

where $\mathcal{L}$ represents the chosen regression objective, typically the Mean Squared Error (MSE) or Huber loss, which mitigates the influence of outlier topological states during optimization. By minimizing this objective, the network intrinsically learns to untangle the highly multimodal spatial features mapping to the specific energetic balances of the extended Heisenberg Hamiltonian.

To achieve robust feature extraction across varying scales of magnetic textures (e.g., from localized domain walls to global skyrmion geometries), this framework evaluates a conventional convolutional baseline alongside four state-of-the-art deep learning architectures: ResNet50, DenseNet, Xception, and the Visual Transformer (ViT). All models are trained from scratch on the magnetic dataset, without any transfer learning from natural-image collections. The four convolutional backbones (the baseline CNN, ResNet50, DenseNet, and Xception) share the common pipeline illustrated in Figure 2—a convolutional feature extractor followed by global average pooling and the regression head—and differ only in the internal structure of their convolutional block, formalized in the equations below. The ViT departs from this scheme, replacing convolutions with self-attention over image patches (Figure 3).

- *Conventional CNN (baseline)*: To quantify the accuracy attainable without transfer learning or the specialized inductive biases of the architectures below, we include a compact convolutional network trained from scratch directly on the native-resolution s_z maps. It stacks four convolutional stages of increasing width (each comprising two 3×3 convolutions with batch normalization, ReLU activation, and spatial dropout, interleaved with spatial max-pooling), followed by global average pooling and the shared regression head. Lacking residual, dense, depthwise-separable, or attention mechanisms, this baseline isolates the marginal contribution of those components to the inverse estimation and serves as the reference against which the deeper architectures are compared (see Figure 2).

- *Residual Networks (ResNet50)* [43]: Standard deep convolutional networks often suffer from the vanishing gradient problem as depth increases, hindering the learning of complex physical mappings. ResNet50 circumvents this by introducing shortcut connections that bypass multiple convolutional layers, directly propagating spatial information and gradients. Mathematically, if $\mathbf{x}$ is the input to a sequence of layers, and $\mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\})$ represents the residual mapping to be learned (comprising convolutions, batch normalization, and ReLU activations) for the l-th layer, the output $\mathbf{y}$ of the residual block is defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_l\}) + \mathbf{x}. \tag{9}$$

In the context of magnetic domain images, these skip connections allow the network to simultaneously preserve low-level spatial features (like abrupt chiral boundaries) while composing deep, high-level abstractions of the global thermodynamic state.

- *Densely Connected Convolutional Networks (DenseNet)* [44]: While ResNet adds features via summation, DenseNet concatenates them, maximizing information flow and feature reuse throughout the network. This is particularly advantageous for the Hamiltonian inverse problem, as the delicate energetic competition often manifests in subtle, overlapping spatial frequencies that require dense multi-scale analysis. In a DenseNet block, the feature maps of all preceding layers are used as inputs to the current layer. If $\mathbf{x}_l$ denotes the output of the l-th layer, and $H_l(\cdot)$ represents a composite function of batch normalization, ReLU, and convolution, the forward pass is expressed as:

$$\mathbf{x}_l = H_l([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{l-1}]), \tag{10}$$

where $[\ldots]$ denotes the concatenation operation. This dense topological structure ensures that the final regression head has direct access to both fine-grained pixel interactions and coarse global structures.

- *Xception (Extreme Inception)* [45]: This architecture operates on the hypothesis that cross-channel correlations and spatial correlations in feature maps can be entirely decoupled. It replaces standard Inception modules with depthwise separable convolutions, drastically reducing computational overhead while enhancing representational efficiency. A depthwise separable convolution splits the standard convolution into two distinct steps. First, a depthwise convolution applies a single spatial filter to each input

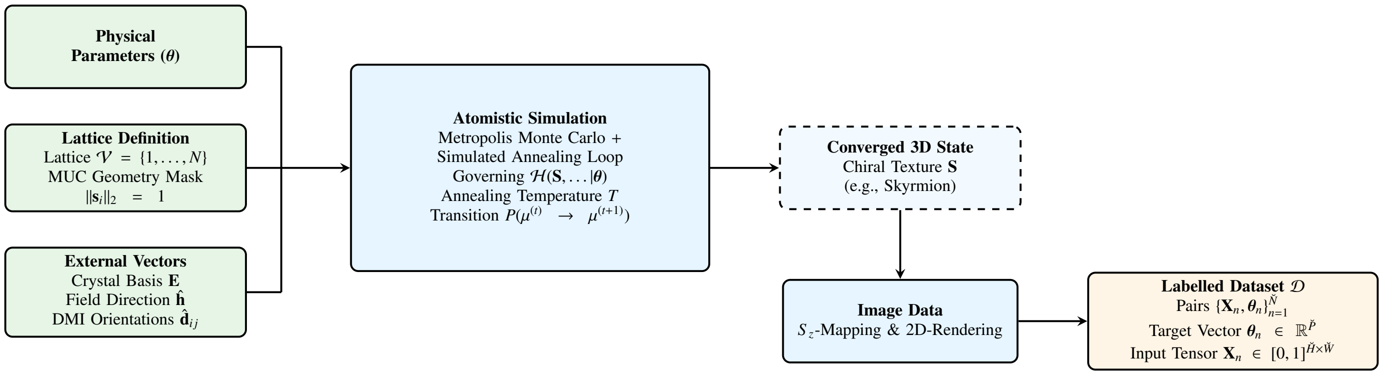

Figure 1: Flowchart of the image-based magnetic dataset generation. Foundational inputs (green)—the Hamiltonian parameters $\boldsymbol{\theta}$, the unit-cell geometry, and the external directional vectors—drive the atomistic Metropolis Monte Carlo simulation with simulated annealing (blue), which relaxes the system into a converged 3D topological state. The resulting vector field is then projected, rendered, and preprocessed (yellow) into the labelled dataset $\mathcal{D}$ mapping 2D s_z images to their generating parameters.

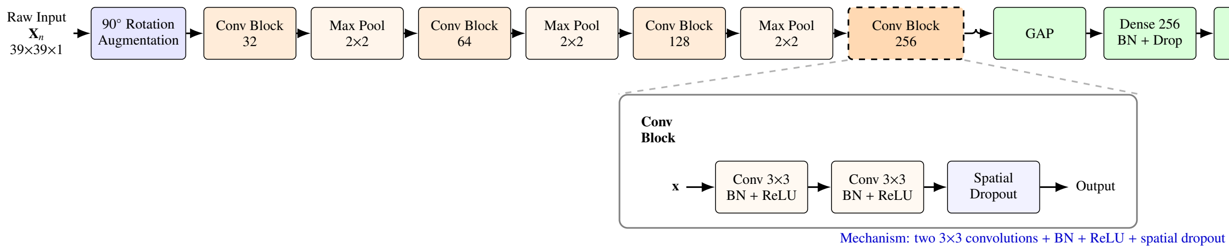

Figure 2: Shared convolutional pipeline used by all four convolutional backbones (baseline CNN, ResNet50, DenseNet, Xception): on-the-fly 90° rotation augmentation, a convolutional feature extractor with spatial max-pooling, Global Average Pooling (GAP), and the regression head. The architectures differ only in the internal convolutional block; the inset shows the baseline block (two 3×3 convolutions with batch normalization, ReLU, and spatial dropout), while the residual, dense, and depthwise-separable variants are formalized in Eqs. 9–11.

channel independently. Second, a pointwise convolution (1×1 convolution) projects the output of the depthwise step onto a new channel space to capture cross-channel correlations. If $\mathbf{X} \in \mathbb{R}^{\check{H} \times \check{W} \times \check{C}}$ is the input tensor, the operation is formally approximated as:

$$\mathrm{Conv}_{\mathrm{sep}}(\mathbf{X}) = \mathrm{Conv}_{\mathrm{pointwise}}(\mathrm{Conv}_{\mathrm{depthwise}}(\mathbf{X})). \tag{11}$$

For nanoscale magnetism, this extreme decoupling allows the network to efficiently isolate specific topological frequency patterns (depthwise) before correlating them to the complex parameter space $\boldsymbol{\theta}_n$.

– *Visual Transformer (ViT)* [46]: Unlike Convolutional Neural Networks, which enforce a strict local inductive bias, the Vision Transformer processes the 2D spatial magnetization field $\mathbf{X}_n$ as a sequence of flattened, non-overlapping 2D patches. This allows the model to capture long-range, global dependencies across the nanodot instantly, which is critical for understanding macro-scale topological stability dictated by the Zeeman field or global anisotropy. The input image is divided into N_p patches, which are linearly projected into a sequence of D-dimensional embeddings. The core mechanism is the Multi-Head Self-Attention (MSA), which computes attention scores to weigh the importance of different patches relative to one another. For a given set of queries Q, keys K, and values V derived from the patch embeddings, the scaled dot-product attention is defined as:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{12}$$

where d_k is the scaling factor (the dimension of the keys). By employing MSA, the ViT can explicitly correlate a localized structural defect on one edge of the nanodot with a compensatory chiral boundary on the opposite edge, mapping these distant correlations directly to the global DMI or symmetric exchange strengths (see Figure 3).

To optimize the parameter set $\hat{\Theta}$ in Equation 8 across all evaluated architectures, the models are trained utilizing the MSE as the explicit continuous loss function $\mathcal{L}$. The optimization process is executed using the Adam optimizer, relying on backpropagation and automatic differentiation to efficiently compute the network gradients [12]. By iteratively updating the model weights based on these computed gradients, the training pipeline systematically minimizes the structural and energetic discrepancies between the predicted outputs and the true Hamiltonian constants, ensuring robust convergence in estimating the underlying physical parameters. All architectures—the conventional baseline and the four deeper models—are trained from scratch under an identical protocol, including the on-the-fly 90° rotation augmentation of the input s_z maps, so that any observed performance difference reflects the architec-

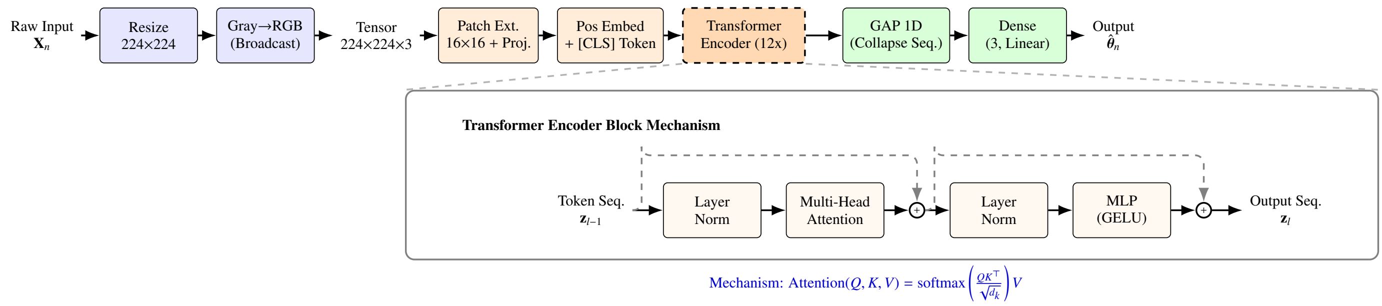

Figure 3: ViT pipeline: patch extraction, linear projection, a backbone of Transformer Encoder blocks, 1D Global Average Pooling, and the regression head. Inset: the Multi-Head Self-Attention (MSA) block modeling long-range spatial dependencies across the nanodot.

ture itself rather than pretraining or data-pipeline discrepancies.

3.4. Spatial Interpretability via RAMs

Interpretability methods based on activation mapping have become a foundational approach for understanding the decision-making processes of deep neural networks. By identifying the spatial regions within an input that most significantly influence a model's prediction, these methods project opaque latent representations back into a human-interpretable domain. Among these, Gradient-weighted Class Activation Mapping (Grad-CAM) and its variants have been extensively utilized in computer vision to generate class-specific saliency maps [47].

In standard classification paradigms, given an input image $\mathbf{X}$ and a discrete target class score y^c, activation maps are computed by extracting the gradients of y^c with respect to the k-th feature map $\mathbf{A}^k \in \mathbb{R}^{H' \times W'}$ of a terminal convolutional layer (where H' and W' denote the spatial dimensions of the feature map). The spatial importance weights are derived by globally pooling these gradients, and the final map is a ReLU-activated linear combination of the feature maps [48]. While highly effective for discrete semantic categorization, this standard formulation structurally fails in continuous regression tasks. The absence of a discrete target class removes the natural selection mechanism required to isolate the output dimension driving the interpretability process. Furthermore, directly computing gradients with respect to a prediction $\hat{\boldsymbol{\theta}}$ introduces severe gradient instability; minute fluctuations in the scalar output can yield drastically different spatial attributions, failing to provide the physical consistency required for analyzing complex Hamiltonian parameters [49].

3.4.1. RAMs for CNN-based Approaches

To overcome the fundamental limitations of discrete activation methods in highly degenerate physical systems, we introduce RAMs, a novel formulation explicitly tailored to bridge continuous parameter estimation with spatial topologies. Given an input magnetic domain visualization $\mathbf{X}$ and the optimized regression model $f_{\hat{\Theta}} : \mathbb{R}^{\check{H} \times \check{W}} \to \mathbb{R}^{\check{P}}$, the network predicts a vector of physical constants $\hat{\boldsymbol{\theta}} = f_{\hat{\Theta}}(\mathbf{X})$. Let $\hat{\theta}_p \in \hat{\boldsymbol{\theta}}$ denote the continuous prediction for the p-th target Hamiltonian parameter (e.g., the DMI strength $\tilde{K}_{DM}$ or symmetric exchange $\tilde{J}$), and let $\theta_p \in \boldsymbol{\theta}$ represents its corresponding ground truth. Instead of utilizing a raw predicted scalar, we formulate a scale-invariant, error-based activation objective O_p. To prevent the mathematical discontinuities and negative objective bounds inherent to absolute error formulations $(1 - |\theta_p - \hat{\theta}_p|)$, we utilize a normalized Gaussian decay function:

$$O_p = \exp\left(-\gamma_p \left(\theta_p - \hat{\theta}_p\right)^2\right), \tag{13}$$

where $\gamma_p > 0$ is a hyperparameter-specific precision factor determining the strictness of the error decay. This objective smoothly bounds the activation target strictly within $(0, 1]$, promoting maximum gradient response precisely when the network's prediction asymptotically approaches the true thermodynamic value, thereby isolating the topological features that positively contribute to accurate regression.

The spatial importance weights $\alpha_k^{(p)}$ for each continuous feature map $\mathbf{A}^k$ are subsequently computed by globally averaging the gradients of the error objective O_p with respect to the local spatial activations $\mathbf{A}_{ij}^k$:

$$\alpha_k^{(p)} = \frac{1}{H'W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \frac{\partial O_p}{\partial \mathbf{A}_{ij}^k}. \tag{14}$$

These weights strictly quantify the importance of the k-th latent spatial feature in successfully minimizing the prediction error for the target physical parameter θ_p. The resulting RAM is then formulated as:

$$\mathcal{R}_{\text{RAM}}^{(p)} = \text{ReLU}\left(\sum_k \alpha_k^{(p)} \mathbf{A}^k\right). \tag{15}$$

The application of the ReLU activation ensures that only spatial features exerting a positive, stabilizing influence on the estimation of the target parameter are preserved. Finally, $\mathcal{R}_{\text{RAM}}^{(p)}$ is spatially upsampled to the original input resolution $\check{H} \times \check{W}$ and normalized to the interval $[0, 1]$. Then, by mapping the continuous error gradients back to the input spatial magnetization field $\mathbf{X}$, RAMs allow for the direct visual and quantitative identification of the specific emergent textures that the network utilizes to deduce the underlying energetic balances. The complete workflow for this localized spatial interpretability is illustrated in Figure 4.

3.4.2. RAMs for ViT-based Approaches

While our standard RAMs formulation seamlessly integrates with CNNs by exploiting their innate spatial hierarchies, ex-

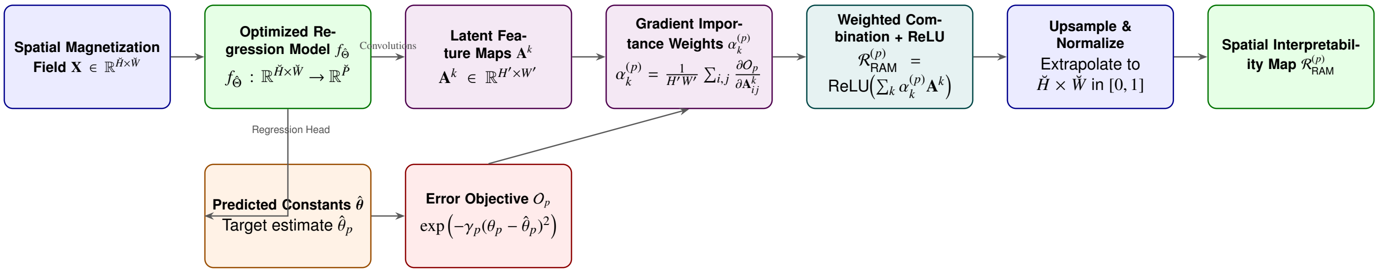

Figure 4: RAMs extraction pipeline. Latent feature maps and continuous predictions from the network backbone are explicitly coupled via an error-decay objective. Gradient-based importance weights combine these features to produce a normalized, physically traceable spatial saliency map for the target Hamiltonian parameter θ_p.

tending this interpretability metric to the Vision Transformers requires mapping sequence-based latent representations back to the two-dimensional physical domain. Unlike CNNs, which preserve spatial locality through sliding convolutional filters, the ViT processes the input spatial magnetization field $\mathbf{X}$ as a flattened sequence of N_r non-overlapping patches, outputting a discrete sequence of latent embeddings. To evaluate ViT-based architectures within the exact same RAMs framework, we introduce a token-reshaping mechanism that bridges the one-dimensional transformer latent space with the two-dimensional topological geometry. Let the output of the terminal Transformer encoder block be denoted as $\mathbf{Z} \in \mathbb{R}^{(N_r+1)\times D}$, where D is the embedding dimension and the extra token corresponds to the global aggregation token (e.g., the `[CLS]` token) utilized by the regression head to predict the Hamiltonian parameters $\hat{\mathbf{y}}$. By isolating the N_r spatial tokens, we obtain the localized sequence $\mathbf{Z}_s \in \mathbb{R}^{N_r\times D}$. Because each sequence token explicitly corresponds to a fixed physical patch of the original nanodot domain, $\mathbf{Z}_s$ can be folded back into a 2D spatial grid. Assuming square patches, the sequence is reshaped into a pseudo-feature map:

$$\mathbf{A}_{\text{ViT}} = \text{Reshape}(\mathbf{Z}_s) \in \mathbb{R}^{\sqrt{N_r}\times\sqrt{N_r}\times D}. \tag{16}$$

Once projected into this tensor format, $\mathbf{A}_{\text{ViT}}$ functions structurally identically to the terminal convolutional feature map $\mathbf{A}$ utilized in the CNN baseline. The k-th slice of the embedding dimension ($k \in \{1, \ldots, D\}$) forms the latent spatial map $\mathbf{A}_{\text{ViT}}^k \in \mathbb{R}^{\sqrt{N_r}\times\sqrt{N_r}}$. We then apply the identical scale-invariant, error-based activation objective O_p defined in Equation 13. The spatial importance weights for the ViT embeddings are computed by globally averaging the gradients of this objective with respect to the reshaped spatial tokens:

$$\alpha_k^{(p)} = \frac{1}{N_r} \sum_{i=1}^{\sqrt{N_r}} \sum_{j=1}^{\sqrt{N_r}} \frac{\partial O_p}{\partial (\mathbf{A}_{\text{ViT}})_{ij}^k}. \tag{17}$$

RAMs for the ViT are constructed via the ReLU-activated linear combination of these reshaped embedding slices, precisely mirroring Equation 15, before being upsampled to the original input resolution $\breve{H} \times \breve{W}$. This extension guarantees an architecture-agnostic evaluation of interpretability, ensuring that the isolated degenerate textures driving the model's predictions can be fairly compared across fundamentally disparate deep learning architectures.

3.4.3. Normalized RAMs for Layer-Wise and Target-Specific Analysis

To rigorously compare the spatial activations across different Hamiltonian parameters and network depths, it is imperative to compute the RAMs utilizing strictly normalized predictions and targets. Let $\bar{\theta}_p \in [0, 1]$ and $\hat{\bar{\theta}}_p \in [0, 1]$ denote the Min-Max normalized ground truth and predicted values for the p-th physical parameter, respectively. By scaling the variables to a uniform interval, the layer-specific error objective is strictly bounded and evaluated as $O_p = \exp(-\gamma_p(\bar{\theta}_p - \hat{\bar{\theta}}_p)^2)$. For a given convolutional or transformer layer $l \in \{1, \ldots, L\}$, we extract the layer-specific unnormalized RAM, denoted as $\mathcal{R}_{\text{RAM}}^{(l,p)}$, following the gradient and activation protocols established in Equations 14 and 15. To quantitatively evaluate and isolate the relative contribution of each distinct layer to the estimation of every physical constant, we introduce a global cross-layer and cross-parameter normalization scheme. The globally normalized activation map, $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)} \in [0, 1]$, is defined by scaling the layer-specific maps against the absolute maximum spatial activation observed across all L layers and all $\breve{P}$ parameters:

$$\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)} = \frac{\mathcal{R}_{\text{RAM}}^{(l,p)}}{\max\limits_{l'\in\{1,\ldots,L\},p'\in\{1,\ldots,\breve{P}\}} \left(\max\limits_{i,j} \left[\mathcal{R}_{\text{RAM}}^{(l',p')} \right]_{i,j} \right)}. \tag{18}$$

This comprehensive normalization ensures that the gradient magnitudes are dimensionally consistent and comparable, preventing parameters with intrinsically higher variance or layers with larger gradient flows from disproportionately dominating the interpretability metrics. By tracking the evolution of $\tilde{\mathcal{R}}_{\text{RAM}}^{(l,p)}$ across the network depth, this formulation explicitly reveals how the deep learning architecture bypasses the intrinsically ill-conditioned nature of the inverse problem. In magnetic nanodots, severe structural ambiguity arises because disparate energetic combinations of the symmetric exchange ($\tilde{J}$), DMI ($\tilde{K}_{DM}$), magnetocrystalline anisotropy ($\tilde{K}$), and Zeeman field ($\tilde{H}_{ex}$) can yield highly degenerate states with nearly identical free energies. Then, the global normalization aim to code how the network dynamically resolves this degeneracy through a hierarchical physical decoupling. We establish that early network layers (lower l) exhibit high relative activations for parameters governing short-range spin interactions, capturing the localized chiral winding angles and domain wall boundaries dictated primarily by the delicate competition between $\tilde{J}$ and $\tilde{K}_{DM}$. Conversely, deeper layers (higher l) aggregate these localized topological textures to resolve macro-scale thermodynamic stability, with the terminal layers concentrating their activations on the globally-coupled parameters—most notably the annealing temperature $T^{(0)}$ and the exchange and field terms that govern the overall degree of order, phase transitions, and spatial alignment of emergent structures such as skyrmion cores and labyrinthine patterns. By correlating the intensity and spatial localization of $\tilde{\mathcal{R}}^{(l,p)}_{\text{RAM}}$ across the network hierarchy, our framework utilizes both localized boundary chirality and global domain spacing to disambiguate the underlying Hamiltonian parameters.

3.4.4. Perturbation-Based Attribution Baselines

To rigorously assess the RAMs formulation, we benchmark it against three model-agnostic attribution methods adapted here to the continuous regression setting: the Local Interpretable Model-agnostic Explanations (LIME), the Shapley Additive Explanations (SHAP) in their kernel form over superpixels, and a spatial-frequency Shapley estimator (SF-SHAP) that evaluates exact Shapley values over a compact set of radial frequency bands. For a target parameter p, let $f_p : \mathbb{R}^{\breve{H}\times\breve{W}} \to \mathbb{R}$ denote the scalar map returning the p-th predicted parameter, $f_p(\mathbf{X}) = \hat{\theta}_p$. The input image is partitioned into d disjoint interpretable regions (superpixels) $\{\mathcal{X}_1, \ldots, \mathcal{X}_d\}$ obtained by spatial segmentation. A coalition vector $\mathbf{z} \in \{0,1\}^d$ encodes which regions are retained ($z_i = 1$) or masked to a reference baseline b ($z_i = 0$), and the map $h_{\mathbf{X}}(\mathbf{z})$ reconstructs the corresponding perturbed image.

LIME. This method fits a sparse linear surrogate $g(\mathbf{z}) = \phi_0 + \sum_{i=1}^{d} \phi_i z_i$ that locally approximates f_p in the neighborhood of $\mathbf{X}$, by minimizing a proximity-weighted least-squares objective:

$$\boldsymbol{\phi}^{\text{LIME}} = \arg\min_{g} \sum_{\mathbf{z}} \pi_{\mathbf{X}}(\mathbf{z}) \left(f_p(h_{\mathbf{X}}(\mathbf{z})) - g(\mathbf{z})\right)^2 + \Omega(g), \quad (19)$$

where $\pi_{\mathbf{X}}(\mathbf{z})$ is a proximity kernel that weights each perturbation by its similarity to the original image, and $\Omega(g)$ penalizes surrogate complexity to enforce sparsity. The coefficient ϕ_i is taken as the attribution of region i.

SHAP. Grounded in cooperative game theory, the Shapley value attributes to each region i the average marginal contribution it makes to the prediction across all possible coalitions:

$$\phi_i = \sum_{\mathcal{S}\subseteq\mathcal{I}\setminus\{i\}} \frac{|\mathcal{S}|!\,(d-|\mathcal{S}|-1)!}{d!} \Big(f_p(h_{\mathbf{X}}(\mathbf{z}_{\mathcal{S}\cup\{i\}})) - f_p(h_{\mathbf{X}}(\mathbf{z}_{\mathcal{S}}))\Big), \quad (20)$$

where $\mathcal{I} = \{1, \ldots, d\}$ and $\mathbf{z}_{\mathcal{S}}$ is the coalition activating only the regions in $\mathcal{S}$. This attribution is the unique one satisfying local accuracy, missingness, and consistency, but its exact evaluation requires 2^d model queries. Since d equals the number of superpixels, exact enumeration is intractable, and SHAP therefore estimates the Shapley values through the kernel formulation—a weighted linear regression over sampled coalitions using the Shapley kernel

$$\pi_{\text{SHAP}}(\mathbf{z}) = \frac{d-1}{\binom{d}{|\mathbf{z}|}\,|\mathbf{z}|\,(d-|\mathbf{z}|)}, \quad (21)$$

so that $\boldsymbol{\phi}^{\text{SHAP}} = \arg\min_{\boldsymbol{\phi}} \sum_{\mathbf{z}} \pi_{\text{SHAP}}(\mathbf{z})(f_p(h_{\mathbf{X}}(\mathbf{z})) - \phi_0 - \sum_i \phi_i z_i)^2$; the resulting coefficients converge to the Shapley values of Equation 20 while sampling only a subset of coalitions.

SF-SHAP. Rather than segmenting the image into superpixels, this variant partitions it into a small number Q of concentric radial frequency bands, matched to the characteristic spatial scales of the magnetic texture. Because Q is small ($Q = 7$ here), the full coalition space of $2^Q = 128$ subsets can be enumerated exactly, so SF-SHAP evaluates the Shapley values of Equation 20 without sampling approximation, at the cost of a coarser, frequency-resolved (rather than pixel-resolved) attribution.

Crucially, all three baselines are perturbation-based: each explanation demands hundreds to thousands of forward evaluations of f_p to probe the coalition space, whereas a RAM is obtained from a single backward pass through the network.

To quantitatively and fairly compare the spatial faithfulness of every attribution method—RAMs and the perturbation baselines alike—we adopt a deletion protocol based on the *Most Relevant First* (MoRF) criterion. Given the relevance map produced by a method m for the parameter p, the $\breve{H}\breve{W}$ input locations are ranked in order of decreasing relevance, yielding the ordering $\pi_1^{(m)}, \pi_2^{(m)}, \ldots, \pi_{\breve{H}\breve{W}}^{(m)}$. For a deletion fraction $\rho \in [0, \rho_{\max}]$, the $\lfloor \rho\, \breve{H}\breve{W} \rfloor$ most relevant locations are replaced by the reference baseline b, producing the perturbed image $\mathbf{X}^{(m)}(\rho)$, and the induced deviation of the prediction is measured as

$$\Delta_p^{(m)}(\rho) = \left| f_p(\mathbf{X}^{(m)}(\rho)) - f_p(\mathbf{X}) \right|. \quad (22)$$

A faithful attribution concentrates relevance on the locations that genuinely drive the prediction, so that removing them induces a large and rapid deviation. The overall faithfulness of method m for parameter p is therefore summarized by the area under the deletion curve,

$$\text{AUC}_p^{(m)} = \int_0^{\rho_{\max}} \Delta_p^{(m)}(\rho)\, d\rho, \quad (23)$$

where a larger $\text{AUC}_p^{(m)}$ indicates a more spatially localized and physically faithful explanation; a random-relevance map is included as a lower-bound control. This deletion score, together with the per-explanation computational cost, provides the quantitative basis for the interpretability comparison reported in Section 5.

4. Experimental Set-Up

To evaluate the effectiveness of the proposed framework for Hamiltonian parameter estimation, we design a comprehensive experimental protocol that integrates physics-based simulations, data-driven modeling, and interpretability analysis. The objective of this setup is to systematically assess not only the predictive performance of the model, but also its ability to generalize across different physical regimes and provide meaningful spatial explanations through the proposed RAMs. The dataset utilized in this study consists entirely of simulated magnetic domain configurations generated through our atomistic

Monte Carlo framework, as described in Section 3.2. To accurately reflect the finite-size effects and surface boundaries inherent to experimental nanostructures, the magnetic nanodots were modeled on a three-dimensional discrete lattice. The cylindrical geometry was constrained by a radius of $R_d = 18.25$ [MUC] and a thickness of $\check{L} = 5$ [MUC]. Within this confined grid, the spatial bounds were dictated by a strictly defined circular binary mask, yielding an active interacting volume of approximately 5,200 localized spins per simulated dot.

For each unique combination of Hamiltonian parameters, the system was subjected to a rigorous simulated annealing protocol to isolate the thermodynamically relaxed state. To guarantee full ergodicity, the exchange-normalized initial temperature was established at an effective paramagnetic ceiling. The system was systematically cooled to a near-ground state of $T^{(f)} = 0.1$ K utilizing a linear decrement of $\Delta T = -0.1$ K. At each discrete temperature interval, the spin lattice underwent 10,000 Monte Carlo Sweeps (MCS) exclusively dedicated to thermalization, thereby overcoming metastable localized energetic barriers. This thermalization phase was subsequently followed by 2,000 measurement MCS to compute macroscopic thermodynamic averages (e.g., specific heat and magnetic susceptibility) and verify strict energetic convergence. To overcome the prohibitive computational bottleneck traditionally associated with sequential Monte Carlo methods, the atomistic simulation was rigorously optimized implemented in JAX as a fully vectorized, just-in-time-compiled routine distributed across multiple GPUs via single-program multiple-data (`pmap`) parallelism in single precision.

Upon converging to a low-energy thermodynamically stable state at 0.1 K, the final topological state was isolated. To mimic experimental 2D observation techniques, the spatial magnetization field was extracted exclusively from the central discrete layer of the nanodot ($z = \lfloor \check{L}/2 \rfloor$). The out-of-plane spin components ($s_z \in [-1, 1]$) were systematically normalized to the strict interval $[0, 1]$ and mapped onto a Cartesian pixel grid, finalizing the input images $\mathbf{X}$ corresponding to the specific physical targets $\boldsymbol{\theta}$. The extended Hamiltonian parameter vector $\boldsymbol{\theta} \in \mathbb{R}^8$ estimated in this study is defined as: $\boldsymbol{\theta} = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$, where the first-shell exchange constant $\tilde{J}_1 = 1.0$ meV is fixed as the energy reference scale to break the scale symmetry of the inverse problem and ensure a well-posed regression target, as described in Section 3.1. Each of the eight free parameters was drawn with a quasi-random (low-discrepancy) sampling scheme within the physically motivated bounds summarized in Table 2, ensuring a systematic and homogeneous coverage of the admissible parameter region.

It is important to note that while all eight parameters are simultaneously estimated by the regression models, their effective identifiability from the s_z projection varies substantially due to the physical constraints of the observational modality. Six of the eight parameters—the annealing temperature $T^{(0)}$, the second-shell exchange $\tilde{J}_2$, the DMI strength $\tilde{K}_{DM}$, the bulk and surface anisotropy constants $\tilde{K}_{an1}$ and $\tilde{K}_{anS}$, and the external field $\tilde{H}_{ex}$—are effectively recovered from the two-dimensional projection. This is made possible by the deliberate enrichment of the dataset: applying the external field along the out-of-plane axis in part of the simulations couples the Zeeman term directly to the observable s_z component, while the extended anisotropy range produces sufficiently distinct domain-wall morphologies for the anisotropy constants to leave a measurable spatial signature. In contrast, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ remain essentially unidentifiable: their energetic contributions are strongly dominated by the first- and second-shell interactions $\tilde{J}_1$ and $\tilde{J}_2$, rendering their individual spatial signatures indistinguishable in the magnetization texture. These two constants are nevertheless retained in the regression target $\boldsymbol{\theta}$ to provide a complete and physically honest characterization of the inverse problem, explicitly delineating its fundamental identifiability limits rather than concealing them. Their inclusion does not degrade the accuracy of the six recoverable parameters, since the multi-output regression head learns to allocate gradient capacity independently to each output dimension. The resulting dataset is publicly available and can be accessed through the Kaggle repository[1]. It contains a total of 166,141 samples, where each input is a grayscale magnetic domain image of size $39 \times 39 \times 1$ pixels, and each corresponding target is a vector of eight physical parameters $\boldsymbol{\theta} \in \mathbb{R}^8$ as defined above.

To broaden the structural diversity of the dataset beyond a single field geometry, the external field orientation was varied within the dataset: in addition to the in-plane configuration, an out-of-plane field was employed to stabilize additional topologies such as isolated skyrmions, while the magnetocrystalline anisotropy range was deliberately extended toward high values to access an Ising-like regime characterized by sharper, more discrete domain walls. The dataset also spans two complementary mechanisms for stabilizing modulated, non-collinear textures: a chiral regime driven by the DMI $\tilde{K}_{DM}$ and an exchange-frustration regime driven by the second-shell exchange $\tilde{J}_2$. To isolate the contribution of each mechanism, these two interactions are activated in a mutually exclusive fashion across the sampled configurations. Consequently, in the pairwise correlation matrix of the eight parameters, $\tilde{J}_2$ and $\tilde{K}_{DM}$ are the only pair exhibiting an appreciable anti-correlation ($\rho \approx -0.53$)—a direct consequence of this deliberate design—while all remaining pairs are essentially uncorrelated ($|\rho| \lesssim 0.3$), confirming that the model cannot exploit spurious cross-parameter correlations to infer one parameter from another. This controlled heterogeneity enriches the space of magnetic phases that the regression models must disentangle and, together with the sampling bounds of Table 2, exposes the models to a wide range of competing energetic regimes within a single unified dataset.

The sampling bounds of Table 2 are not intended to reproduce a single specific compound but to cover the region of Hamiltonian space where chiral and modulated magnetic textures exist. Since $\tilde{J}_1$ is fixed as the energy reference, the physically meaningful quantities are the dimensionless ratios $\tilde{J}_r/\tilde{J}_1$, $\tilde{K}_{DM}/\tilde{J}_1$, and $\tilde{K}/\tilde{J}_1$, and these ratios fall within the ranges reported by first-principles calculations and experiments for chi-

[1] `https://www.kaggle.com/datasets/carloscanamejoy/dataset-spines-complete`

ral magnets and ultrathin transition-metal films. The higher-order exchange constants ($|\tilde{J}_2| > |\tilde{J}_3| > |\tilde{J}_4|$, with $\tilde{J}_2/\tilde{J}_1$ up to 0.66) reproduce the decaying, sign-alternating exchange obtained from density-functional calculations of ultrathin films such as Fe/Ir(111) and Pd/Fe/Ir(111) [53, 54], and lie in the strong-frustration regime of the classical $\tilde{J}_1$–$\tilde{J}_2$ model. For the DMI, we sample $\tilde{K}_{DM}/\tilde{J}_1$ up to ≈ 1.2; while interfacial materials typically exhibit $\tilde{K}_{DM}/\tilde{J}_1 \sim 0.3$–$0.7$ [53], larger ratios are the standard choice in finite-lattice Monte Carlo studies so that the helical period ($\lambda \propto \tilde{J}_1/\tilde{K}_{DM}$) fits several times within the simulation cell [55]. Because the physics is scale-invariant, a large ratio on a small lattice is equivalent to a small ratio in a large system, so this choice reflects standard finite-size rescaling rather than unphysical parameter values. The anisotropy range (up to $\tilde{K}_{an1} \approx 4.6$ meV/atom) spans from soft magnets to the strongly anisotropic interfacial regime, consistent with values reported for systems ranging from FePt to single Co adatoms on Pt(111) [56, 57]. Finally, the Zeeman range ($\tilde{H}_{ex}$ up to 1.2 meV/atom) corresponds, through $E = g\mu_B B$ with $g \approx 2$, to magnetic fields up to ≈ 10 T—the upper range of standard laboratory magnets used to map skyrmion phase diagrams—and the temperature range ($T^{(0)}$ up to 20 K, i.e., $k_B T/\tilde{J}_1$ up to ≈ 1.7) sweeps the system from the ordered ground state through the paramagnetic transition, consistent with ordering temperatures of chiral magnets with $\tilde{J}_1 \sim 1$ meV.

Table 2: Sampling ranges for the eight free Hamiltonian parameters constituting the regression target $\boldsymbol{\theta}$. The first-shell exchange $\tilde{J}_1 = 1.0$ meV is fixed as the energy reference. Parameters are drawn with a quasi-random (low-discrepancy) scheme within the specified bounds. Physical units: temperature in Kelvin [K], exchange and DMI constants in meV, anisotropy constants in meV/atom, external field in meV/atom.

Parameter	**Physical role**	**Min**	**Max**
$T^{(0)}$ [K]	Annealing temperature	0.0	20.0
$\tilde{J}_2$ [meV]	2nd-shell exchange	−0.33	0.66
$\tilde{J}_3$ [meV]	3rd-shell exchange	−0.29	0.29
$\tilde{J}_4$ [meV]	4th-shell exchange	−0.23	0.24
$\tilde{K}_{an1}$ [meV/atom]	Bulk anisotropy	0.0	4.57
$\tilde{K}_{anS}$ [meV/atom]	Surface anisotropy	0.0	0.20
$\tilde{H}_{ex}$ [meV/atom]	External Zeeman field	0.0	1.20
$\tilde{K}_{DM}$ [meV]	DMI strength	0.0	1.20

It is important to note that the DMI strength $\tilde{K}_{DM}$ is sampled exclusively over non-negative values ($\tilde{K}_{DM} \in [0.0, 1.2]$ meV). This restriction is a physically motivated convention arising from the symmetry properties of the antisymmetric exchange term in Equation 4. Under the transformation $\tilde{K}_{DM} \to -\tilde{K}_{DM}$, the equilibrium spin configuration undergoes a reversal of its in-plane chirality—the rotational sense of the chiral winding inverts—while the out-of-plane magnetization component s_z at every lattice site remains strictly invariant. Since the regression models operate exclusively on the two-dimensional s_z projection, configurations with $+\tilde{K}_{DM}$ and $-\tilde{K}_{DM}$ are exactly degenerate in the observation space, rendering the sign of the DMI fundamentally unidentifiable from s_z images alone. Including both signs simultaneously would introduce a strict two-fold degeneracy in the input–output mapping, collapsing the model's predictions toward $\hat{\tilde{K}}_{DM} \approx 0$ and precluding meaningful regression. The chirality handedness is instead encoded through the fixed direction vector $\hat{\mathbf{d}}_{ij}$, consistent with standard conventions in atomistic modeling of interfacial DMI, where the sign of the antisymmetric exchange sets the preferred rotational sense of the chiral texture [39, 58, 52]. Recovering the DMI sign would require access to chirality-sensitive observables, such as the full three-dimensional spin vector field or in-plane magnetization components, which constitutes a direction for future investigation (Section 6).

To ensure robust evaluation and prevent information leakage during the deep learning optimization phase, the generated dataset $\mathcal{D}$ is partitioned into training, validation, and testing subsets using a rigorous two-stage random split strategy. Initially, 15% of the total dataset is isolated as a hold-out test set to evaluate ultimate model generalization. Subsequently, the remaining 85% is further divided, allocating exactly 15% of the total initial data for validation. This procedure yields an effective split of 70% for training, 15% for validation, and 15% for testing. All partitions are executed utilizing a fixed random seed to guarantee strict reproducibility across all experimental runs. This strategy ensures that topological samples remain strictly independent across subsets while perfectly preserving the underlying thermodynamic distributions of the dataset.

Prior to network ingestion, both the input spatial magnetization images and the target Hamiltonian variables are preprocessed to ensure numerical stability and efficient gradient convergence. The physical target variables are scaled using Min-Max normalization, mapping each continuous parameter to the strict interval $[0, 1]$ based exclusively on the statistical distribution of the training set. This identical transformation is subsequently applied to the validation and test sets to maintain inferential consistency and directly support the bounded RAMs objective $O_p \in (0, 1]$. For the four deep backbone architectures, the single-channel s_z matrices are spatially resized to 224×224 pixels and duplicated across three channels (grayscale-to-RGB conversion). This dimensional expansion matches the input dimensionality expected by the standard deep backbone architectures; the conventional CNN baseline instead operates directly on the native $39 \times 39 \times 1$ maps. No pretrained weights are used, and all architectures are trained entirely from scratch.

The configuration of the evaluated deep learning architectures (see Section 3.3) is summarized in Table 3. All five architectures share the same extended regression head designed to mitigate overfitting in the high-dimensional parameter space. Specifically, the global pooling output is passed through a Batch Normalization layer, followed by a Dropout layer with rate $p = 0.4$, a fully connected Dense layer of 256 units with ReLU activation and ℓ_2 weight regularization ($\lambda = 10^{-4}$), a second Batch Normalization layer, a second Dropout layer with rate $p = 0.3$, and a final linear Dense layer outputting the $\check{P} = 8$ target parameters. For the convolutional architectures (the conventional CNN, ResNet50, DenseNet121, and Xception), the global aggregation is performed via Global Average Pooling over the spatial feature maps, whereas for the ViT-B/16 the token sequence output of the terminal encoder block is first collapsed via Global Average Pooling along the sequence dimension before entering the shared regularization head. This design choice reflects the distinct spatial inductive biases of convolutional and attention-based backbones while ensuring a strictly comparable regression capacity across all evaluated models.

The optimization of the parameter set $\hat{\Theta}$ was executed utilizing the Adam optimizer with a unified initial learning rate of $\eta = 10^{-4}$ for all architectures, including the ViT-B/16, chosen for stable convergence when training all architectures from scratch. All models were trained to minimize the MSE loss function, while the MAE was actively monitored as a secondary diagnostic. Training was conducted using a highly parallelized batch size of 256 samples over a maximum of 100 epochs. To actively prevent overfitting and manage the learning dynamics within the highly degenerate optimization landscape, two primary callbacks were implemented: a dynamic learning rate scheduler (`ReduceLROnPlateau`) that reduced the learning rate by a factor of 0.3 if the validation loss stagnated for 5 consecutive epochs, and an `EarlyStopping` protocol that halted training and restored the best network weights if no validation improvement was observed after 12 consecutive epochs.

The predictive performance on the unseen test set was quantitatively assessed using two complementary regression metrics. The coefficient of determination (R^2) quantifies the proportion of variance in each physical parameter successfully captured by the network:

$$R^2 = 1 - \frac{\sum_{n=1}^{N_{\text{test}}} \left(\theta_{n,p} - \hat{\theta}_{n,p}\right)^2}{\sum_{n=1}^{N_{\text{test}}} \left(\theta_{n,p} - \bar{\theta}_{p}\right)^2}, \tag{24}$$

where $\bar{\theta}_p$ is the mean of the observed ground truth target. Note that R^2 is statistically undefined when the target variable exhibits near-zero variance within a given evaluation subset; in these cases the metric is reported as N/A. The Mean Absolute Error (MAE) evaluates the absolute magnitude of the prediction error in the normalized feature space $[0, 1]$, providing a direct and interpretable measure of per-parameter accuracy:

$$\text{MAE} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} \left|\theta_{n,p} - \hat{\theta}_{n,p}\right|. \tag{25}$$

Since all target variables are Min-Max normalized to $[0, 1]$ prior to training, the MAE is inherently dimensionless and directly comparable across all eight Hamiltonian parameters, irrespective of their original physical units or numerical ranges.

The RAMs were computed on the held-out test set using the formulations introduced in Section 3.4. For the Xception backbone, RAMs were extracted from eight layers selected to systematically cover the three functional stages of the architecture: two entry-flow layers capturing low-level spatial features at high resolution — `block1_conv2` (112×112) and `block3_sepconv2` (56×56) —; one transitional layer at `block4_sepconv2` (28×28); five middle-flow layers operating at the bottleneck resolution of 14×14 — `block6_sepconv2`, `block8_sepconv2`, `block10_sepconv2`, and `block12_sepconv2` —; and one exit-flow layer, `block13_sepconv2` (14×14), immediately preceding the global pooling operation. This stratified selection was motivated by the physical hypothesis that early layers encode fine-grained domain periodicity and local spin-winding patterns at the scale of individual domain walls, while middle-flow layers progressively aggregate these local features into intermediate topological representations, and exit-flow layers encode high-level semantic information about the global magnetic phase. By distributing the selected layers across all three functional stages, the analysis directly tests whether the hierarchical decoupling predicted by the normalized RAM formulation of Equation 18 is empirically realized in the Xception backbone for Hamiltonian parameter estimation. For the ViT, a single attribution map per parameter was derived from the spatial token embeddings at the output of the terminal encoder block, reshaped into $\mathbf{A}_{\text{ViT}} \in \mathbb{R}^{14\times14\times768}$ following Equation 16. Since `keras_hub.ViTBackbone` does not expose internal token tensors as differentiable nodes within `tf.GradientTape`, the token sequence $\mathbf{Z}_s \in \mathbb{R}^{196\times768}$ was first extracted via a standard forward pass and then re-instantiated as a `tf.Variable` to enable gradient computation. The `[CLS]` token was treated as a fixed constant and the reconstructed sequence was passed through the `GlobalAveragePooling1D` and `Dense` regression head to compute O_p and its gradients with respect to $\mathbf{Z}_s$, following Equation 17.

The precision factor was fixed to $\gamma_p = 10$ uniformly for all $p \in \{1, \ldots, 8\}$. Given that all target variables are Min-Max normalized to $[0, 1]$, this value ensures a smooth and well-conditioned gradient signal across the full prediction error range: $O_p \geq e^{-10} \approx 4.5 \times 10^{-5}$ at maximum error and $O_p = 1$ at perfect prediction. Lower values of γ_p would yield spatially diffuse attributions insensitive to prediction quality, while higher values would cause gradient saturation for the less identifiable parameters where errors are systematically larger. The chosen value maintains sensitivity across all eight parameters, from the most identifiable ($T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$) to the physically degenerate ones ($\tilde{K}_{an1}$, $\tilde{K}_{anS}$, $\tilde{H}_{ex}$).

To confine activations to the physical extent of the nanodot, a circular binary mask $\mathcal{M} \in \{0, 1\}^{224\times224}$ — defined as a disk of radius $r = 0.94 \times 112$ pixels centered at the image center — was applied to each RAM before normalization, setting all activations outside the disk to zero. A concentration score $\xi \in [0, 1]$, measuring the fraction of total activation energy within the disk, was used to select the most physically representative sample per magnetic phase: among the ten test images closest to the UMAP centroid of each phase cluster, the sample maximizing $\bar{\xi}$ averaged over all parameters and layers was chosen as the phase representative $\mathbf{X}_n^*$.

All experiments were conducted in a cloud environment on RunPod. The computational setup was equipped with an NVIDIA H200 GPU. The training pipeline was implemented in Python using TensorFlow, leveraging GPU acceleration for efficient model training and inference. The atomistic state simulations were executed separately on the multi-GPU JAX configuration described in Section 3.2. The complete source code is publicly available at `https://github.com/deathperminut/PaperInverseProblemEstimation`, organized into three modules: `/simulation` containing the atomistic Monte Carlo Hamiltonian simulation scripts implemented in JAX with multi-GPU (`pmap`) parallelization; `/regression`

Table 3: Configuration of evaluated deep learning architectures for inverse Hamiltonian parameter estimation. All models share a standardized regularized regression head mapped to the $\breve{P} = 8$-dimensional physical parameter space. BN: Batch Normalization; Drop(p): Dropout with rate p.

Model	Architecture Type	Pretraining	Global Aggregation	Output Head
Conventional CNN	Custom CNN	None (from scratch)	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
ResNet50	Residual CNN	None (from scratch)	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
DenseNet121	Densely Connected CNN	None (from scratch)	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
Xception	Depthwise Separable CNN	None (from scratch)	Global Avg. Pooling (spatial)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)
ViT-B/16	Patch-based Transformer	None (from scratch)	Global Avg. Pooling (tokens)	BN → Drop(0.4) → Dense(256, ReLU, ℓ_2) → BN → Drop(0.3) → Dense(8)

containing the training, validation, and evaluation pipelines for all five deep learning architectures; and `/rams` containing the Regression Activation Maps computation and visualization code for both the Xception CNN and the Vision Transformer.

5. Results and Discussion

5.1. Thermodynamic Simulation of Magnetic Topologies

We present the main results of the proposed atomistic Hamiltonian simulation framework (see Section 3). The simulated dataset spans the eight-dimensional parameter space $\boldsymbol{\theta}$ over the bounds summarized in Table 2, sampled with deliberately heterogeneous densities: the higher-order exchange constants $\tilde{J}_2$, $\tilde{J}_3$, and $\tilde{J}_4$ concentrate near zero, whereas the DMI strength $\tilde{K}_{DM}$ is broadly populated across its full [0, 1.2] meV range, and the bulk anisotropy $\tilde{K}_{an1}$ and external field $\tilde{H}_{ex}$ exhibit heavy tails toward high values that deliberately incorporate high-anisotropy, Ising-like regimes and out-of-plane field configurations. These bounded ranges and sampling heterogeneities explicitly govern the topological diversity of the generated magnetic phase space and, in turn, the learning dynamics, optimization stability, and predictive limits of the downstream deep regression frameworks when navigating the challenging regimes of the extended Heisenberg Hamiltonian.

To further validate the physical consistency of the generated dataset and provide an intuitive understanding of the underlying spin dynamics, Figure 5 illustrates the thermodynamic evolution of a representative magnetic nanodot during the Metropolis–Hastings Monte Carlo annealing process. This simulation is conducted under the fixed parameter configuration $\tilde{J}_1 = 1.0$ meV, $\tilde{J}_2 = \tilde{J}_3 = \tilde{J}_4 = 0.0$ meV, $\tilde{K}_{an1} = 0.10$ meV/atom, $\tilde{K}_{anS} = 0.00$ meV/atom, $\tilde{H}_{ex} = 0.00$ meV/atom, and $\tilde{K}_{DM} = 1.00$ meV, which constitutes a representative point within the full eight-dimensional parameter space $\boldsymbol{\theta}$ defined in Table 2. The vanishing values of the higher-order exchange constants ($\tilde{J}_2$, $\tilde{J}_3$, $\tilde{J}_4$), surface anisotropy ($\tilde{K}_{anS}$), and external field ($\tilde{H}_{ex}$) isolate the dominant energetic competition between the first-shell symmetric exchange $\tilde{J}_1$ and the DMI $\tilde{K}_{DM}$, making this configuration particularly illustrative of the chiral ordering mechanism. The main panel tracks the total Hamiltonian energy $\mathcal{H}$ [meV] as a function of the exchange-normalized annealing temperature $T^{(0)}$ [K], while each inset displays the corresponding two-dimensional out-of-plane magnetization field s_z of the central layer at that specific thermal state. At elevated temperatures ($T^{(0)} \gtrsim 15$ K), the spin configurations exhibit a fully disordered, paramagnetic character, manifesting as spatially uncorrelated fluctuations of s_z with no preferred orientation. As the system is progressively cooled through the

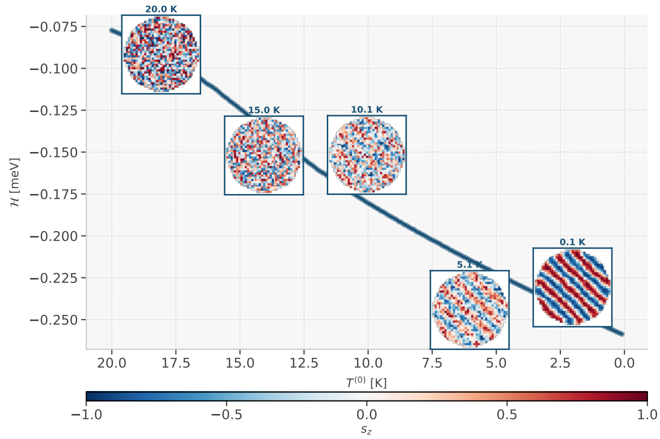

Figure 5: Thermodynamic convergence of a representative nanodot during simulated annealing. The main panel shows the total Hamiltonian energy $\mathcal{H}$ [meV] versus the annealing temperature $T^{(0)}$ [K]; each inset shows the s_z field of the central layer at that state, tracing the transition from a disordered paramagnetic phase at high $T^{(0)}$ to a relaxed labyrinthine configuration at $T^{(0)} = 0.1$ K.

intermediate regime ($5 \text{ K} \lesssim T^{(0)} \lesssim 15$ K), short-range correlated domains begin to nucleate, reflecting the growing dominance of the symmetric exchange interaction $\tilde{J}_1$ over thermal fluctuations. In this regime, the competition between $\tilde{J}_1$ and the $\tilde{K}_{DM}$ progressively induced non-collinear and chiral spin winding, visible as the emergence of alternating red-blue stripe-like patterns. Upon convergence to the near-ground state at $T^{(0)} = 0.1$ K, the system stabilizes into a well-defined labyrinthine domain configuration, characteristic of the helical phase governed by a strong DMI relative to the exchange energy. The monotonic decrease $\mathcal{H}$ throughout the annealing trajectory confirms the thermodynamic relaxation of the system toward a low-energy stable configuration, validating the physical correctness of the Monte Carlo pipeline and the quality of the generated dataset as input for the deep learning regression framework.

To fully characterize the structural diversity generated by the Monte Carlo annealing protocol across the entire dataset, Figure 6 presents two-dimensional phase diagrams of the magnetization image as a function of key Hamiltonian parameter pairs: $T^{(0)}$ versus $\tilde{J}_2$ and $T^{(0)}$ versus $\tilde{K}_{DM}$. These diagrams illustrate the macroscopic topological transitions driven by the underlying atomistic energetic competition. In the low-temperature regime, the system stabilizes into well-defined ordered textures whose morphology is dictated by the parameters: increasing $\tilde{K}_{DM}$ drives the sequence from nearly uniform ferromagnetic states, through chiral stripe and labyrinthine domains, to isolated skyrmionic textures, while the sign of $\tilde{J}_2$ separates ferromagnetic from antiferromagnetically modulated exchange ordering. Crucially, these ordered configurations remain visually distinguishable across the parameter space, which is precisely what allows the majority of the Hamiltonian parameters to be recovered by the inverse models. The principal source of ill-conditioning is instead thermal in origin: as the annealing temperature approaches the effective Curie point T_C, thermal fluctuations progressively overwhelm the magnetic interactions, and the system passes through transient, partially ordered states before collapsing into a paramagnetic, spatially uncorrelated texture that carries little discriminative information about the generating parameters. This temperature-driven loss of structure—together with the weakly discriminative transient regimes, rather than any intrinsic parameter degeneracy—constitutes the dominant challenge the dataset poses to the inverse estimation, consistent with the noisy, ill-conditioned regimes highlighted in the abstract. By comprehensively sampling this phase space, including these transient and high-temperature regimes, the thermodynamic simulation establishes a rich yet rigorously challenging foundation for the subsequent inverse deep learning parameter estimation.

5.2. Deep Learning Regression Results

The predictive performance of the five evaluated architectures (the conventional CNN baseline, DenseNet, ResNet, Xception, and ViT) was assessed on the held-out test set comprising 15% of the total dataset. Each model was tasked with simultaneously inferring the complete extended parameter vector $\boldsymbol{\theta}$ directly from the two-dimensional out-of-plane magnetization image $\mathbf{X}_n$, without any auxiliary physical information. As shown in Figures 8 and 9, all five architectures perform competitively, with the differences on the identifiable parameters typically confined to a few percent of R^2. Among them, Xception emerges as the strongest overall model: it attains the highest R^2 for six of the eight parameters — including the most identifiable ones, $T^{(0)}$, $\tilde{J}_2$, $\tilde{K}_{an1}$, and $\tilde{K}_{DM}$ — while the ViT marginally leads on the external field $\tilde{H}_{ex}$ and the conventional CNN baseline on the surface anisotropy $\tilde{K}_{anS}$. The same ranking is largely mirrored in the MAE panels, where Xception yields the lowest absolute error for the physically dominant parameters $T^{(0)}$, $\tilde{K}_{an1}$, and $\tilde{K}_{DM}$. A particularly informative observation is that the conventional CNN baseline — trained from scratch and devoid of pretraining or specialized inductive biases — matches, and for some parameters even exceeds, the deeper backbones. This confirms that neither transfer learning from natural images nor the sophisticated residual, dense, depthwise, or attention mechanisms provide a decisive advantage for this task; the estimation accuracy is instead governed by the physical information that the s_z projection makes available. Physically, Xception's slight edge stems from its depthwise separable convolution mechanism, which efficiently decouples the extraction of spatial spin-texture frequencies from the cross-channel correlations required to map those features onto the Hamiltonian parameter space, yielding the most accurate point predictions for the dominantly identifiable parameters. The ViT, in turn, remains competitive by processing $\mathbf{X}_n$ as a sequence of non-overlapping spatial patches and computing multi-head self-attention across all patch pairs, explicitly capturing the long-range correlations—such as the global chirality imposed by $\tilde{K}_{DM}$ and the large-scale ordering driven by $\tilde{J}_1$—that are inaccessible to purely local convolutional receptive fields.

This near-equivalence across architectures raises the question of whether model capacity limits the estimation at all. Figure 7 plots the mean R^2 over the identifiable parameters against the number of trainable weights of each architecture, spanning nearly two orders of magnitude in model size—from the compact conventional CNN (~ 1 M parameters) to the ViT (~ 86 M parameters). The resulting trend is essentially flat: the conventional CNN already attains a mean R^2 close to that of the largest models, and increasing the parameter count by two orders of magnitude yields only a marginal improvement, with Xception offering the best accuracy–size trade-off. This saturation demonstrates that predictive accuracy is not bottlenecked by model capacity but by the physical information accessible in the s_z projection: once an architecture is expressive enough to read the relevant spatial features, additional parameters and more elaborate inductive biases provide diminishing returns, confirming that the recovery of the identifiable parameters is a robust, architecture-independent property of the data.

Figures 8 and 9 reveal a pronounced and physically interpretable heterogeneity in predictive accuracy across the parameter space. The initial annealing temperature $T^{(0)}$, the second-shell exchange $\tilde{J}_2$, the DMI strength $\tilde{K}_{DM}$, the bulk anisotropy $\tilde{K}_{an1}$, and the external field $\tilde{H}_{ex}$ consistently yield high R^2 scores across all architectures, reflecting the robust and spatially distinctive topological imprints they leave on the observable magnetization texture: $T^{(0)}$ governs the global degree of spin order, $\tilde{J}_2$ controls the second-neighbor exchange pathway that modulates the domain-scale collinearity, and $\tilde{K}_{DM}$ directly dictates the characteristic wavelength and chirality of the domain patterns. However, a seemingly contradictory observation emerges when examining the MAE panels in Figure 9: despite their high R^2 scores, $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$ also exhibit comparatively larger absolute errors in their respective physical units — reaching up to approximately 0.49 K for $T^{(0)}$, 0.021 meV for $\tilde{J}_2$, and 0.037 meV for $\tilde{K}_{DM}$ in the worst-performing architecture. This apparent inconsistency is not a failure of the models but rather a direct consequence of the thermodynamic landscape of the dataset. As illustrated in the phase diagrams of Figure 6, the parameter space contains extensive high-temperature regions where thermal fluctuations above the effective Curie temperature T_C collapse the magnetization texture into an indistinguishable paramagnetic noise regime, regardless of the specific values of $\tilde{J}_2$ or $\tilde{K}_{DM}$. Within these thermally disordered regimes, configurations generated by widely different parameter values relax into statistically similar noise textures, inflating the absolute prediction error even when the global correlation structure — captured by R^2 — remains strong. The MAE values for $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$ therefore reflect the thermal indistinguishability of the high-temperature paramagnetic regime rather than a systematic modeling deficiency: the same parameter that is most accurately recovered in the ordered low-temperature regime (where the spin texture preserves a clear topological signature) becomes the least constrained in the disordered high-

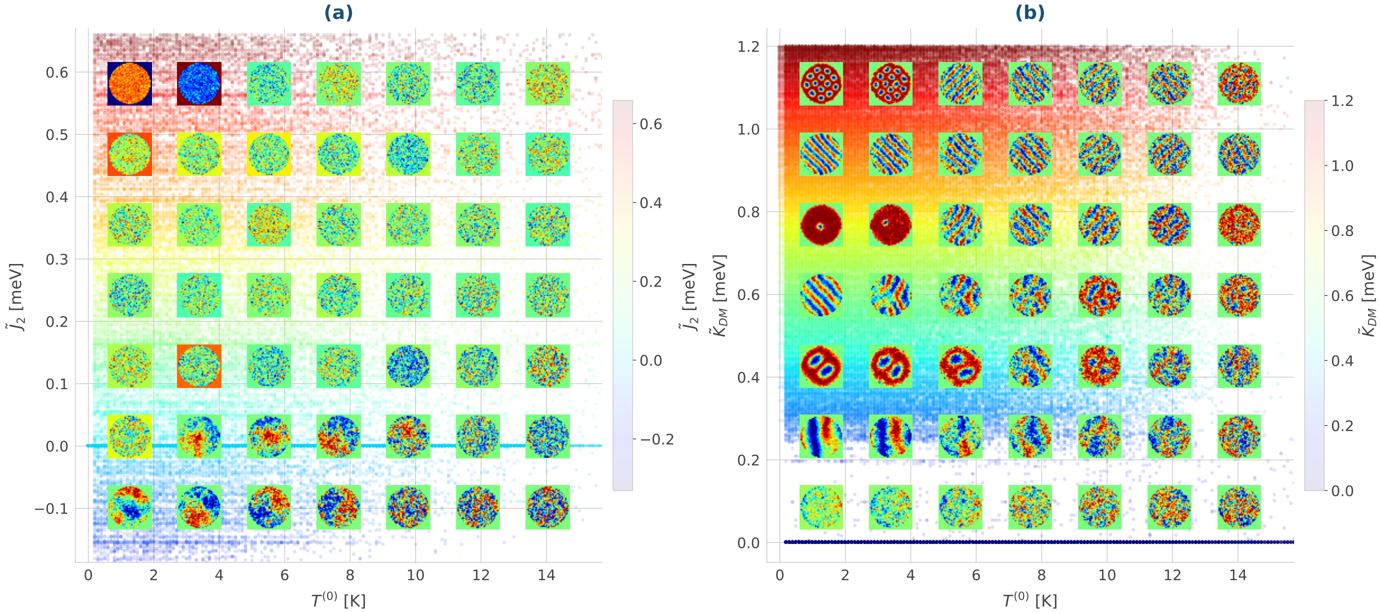

Figure 6: Two-dimensional phase diagrams of the magnetization texture $\mathbf{X}_n$ over parameter pairs: **(a)** $T^{(0)}$ vs $\tilde{J}_2$ and **(b)** $T^{(0)}$ vs $\tilde{K}_{DM}$. Each cell shows an enlarged representative image on a coarse sampling grid; the background color encodes the vertical-axis parameter.

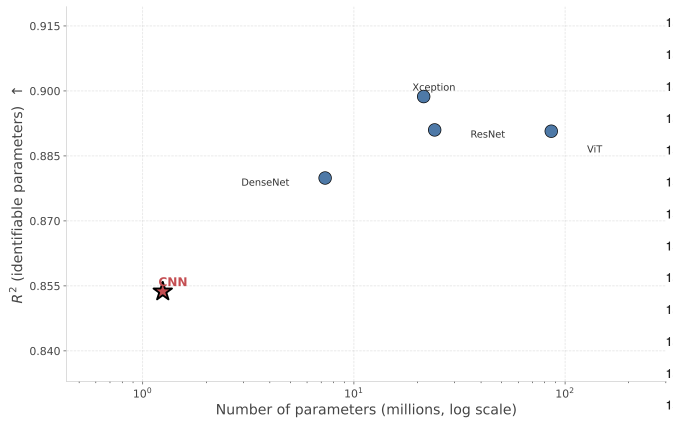

Figure 7: Mean R^2 over the identifiable Hamiltonian parameters versus the number of trainable weights (log scale) for the five architectures. The nearly flat trend—the compact CNN already matching models two orders of magnitude larger—indicates that accuracy is limited by the physical information in the s_z projection, not by model capacity.

temperature regime (where the texture collapses into thermal noise), producing non-negligible global MAE values driven primarily by the ill-conditioned, temperature-dominated subset of the dataset. A rigorous quantitative decomposition of this effect requires first separating the dataset into its constituent magnetic structural categories, as the global aggregation of all configurations inevitably conflates high-accuracy ordered-phase predictions with fundamentally ill-posed paramagnetic-regime estimates. This per-phase analysis is addressed further ahead, where the unsupervised structural clustering of the magnetization images allows us to isolate and quantify the specific contribution of each magnetic phase to the observed prediction error.

Conversely, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ delimit the fundamental identifiability boundary of the inverse problem in the 2D observation space. They constitute the only genuinely unidentifiable parameters: their R^2 scores remain near zero across all evaluated architectures, indicating that the models are unable to recover meaningful information about them beyond the dataset mean. Physically, the spatial signatures of these subordinate exchange constants are energetically dominated by the primary $\tilde{J}_1$ and $\tilde{J}_2$ interactions, masking their individual effect on the magnetization texture and rendering them effectively unidentifiable from the s_z projection alone. Notably, their MAE values — on the order of 0.03 meV — remain numerically comparable to those of the better-identified parameters, which is a direct consequence of their narrow physical range rather than genuine predictive skill: a model defaulting to the dataset mean for $\tilde{J}_3$ and $\tilde{J}_4$ naturally incurs a small absolute error simply because the dynamic range of these parameters is itself limited. In contrast — and in marked departure from what a purely in-plane field geometry would permit — the external Zeeman field $\tilde{H}_{ex}$ and the bulk anisotropy $\tilde{K}_{an1}$ are recovered with high fidelity ($R^2 \gtrsim 0.94$). Applying the field along the out-of-plane axis in part of the dataset couples the Zeeman term directly to the observable s_z component, while the extended anisotropy range imprints sufficiently distinct domain-wall morphologies for $\tilde{K}_{an1}$ to leave a strong, recoverable spatial signature. The surface anisotropy $\tilde{K}_{anS}$ occupies an intermediate regime ($R^2 \approx 0.65$), as its visual manifestation remains partially entangled with the bulk anisotropy $\tilde{K}_{an1}$ and is geometrically confined to the thin boundary layer of the nanodot. Taken together, these results establish that six of the eight Hamiltonian parameters are effectively recovered from the two-dimensional s_z projection, with only the two higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ marking the fundamental identifiability limit of the inverse problem.

To gain deeper insight into the internal representations learned by the best-performing model, Xception, during the inverse Hamiltonian estimation task, Figure 10 presents a Uni-

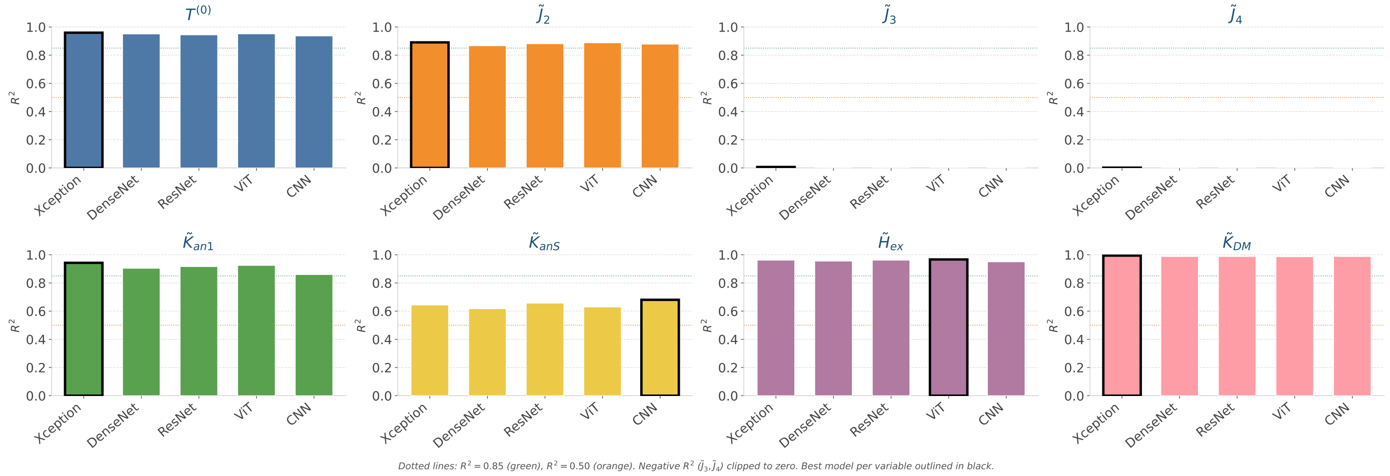

Figure 8: Per-variable R^2 score across the five evaluated architectures (Xception, DenseNet, ResNet, ViT, and the conventional CNN baseline) for the eight Hamiltonian parameters $\boldsymbol{\theta}$ on the held-out test set. Each panel is one parameter, one bar per architecture; negative R^2 values are clipped to zero, and black-bordered bars mark the best architecture per parameter.

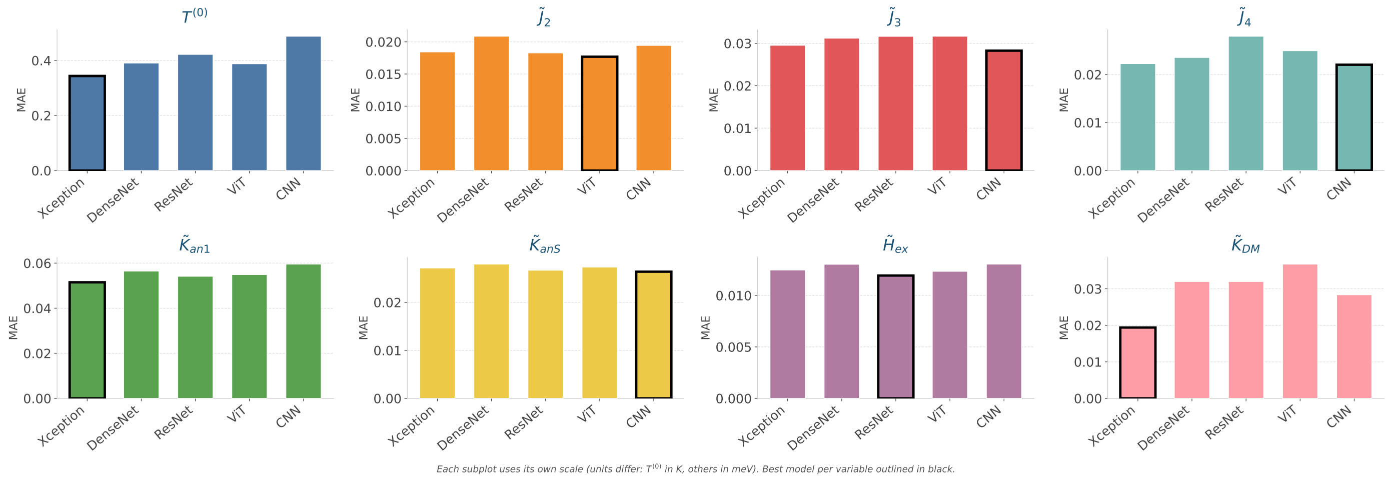

Figure 9: Per-variable Mean Absolute Error (MAE) across the five evaluated architectures for the eight Hamiltonian parameters $\boldsymbol{\theta}$ on the held-out test set, in each parameter's physical units (K for $T^{(0)}$, meV for exchange and DMI, meV/atom for anisotropy and field). Each panel is one parameter, one bar per architecture; black-bordered bars mark the best architecture. The small MAE of $\tilde{J}_3$ and $\tilde{J}_4$ despite their near-zero R^2 reflects their narrow dynamic range rather than genuine predictive skill.

form Manifold Approximation and Projection (UMAP) of the model's latent space [50], where each point corresponds to a test sample colored by the value of each of the eight Hamiltonian parameters $\boldsymbol{\theta}$. The UMAP embedding was computed from the high-dimensional feature vectors extracted from the Xception backbone prior to the regression head, providing a two-dimensional visualization of the geometric structure that the model has learned to impose on the magnetization image manifold. Rather than forming a single undifferentiated cloud, the latent space exhibits a rich topological structure composed of several geometrically distinct clusters and continuous gradients, reflecting the fact that the model has organized the magnetization textures according to their underlying physical regimes. Examining the coloring by $T^{(0)}$ reveals the clearest and most spatially coherent gradient across the embedding: low-temperature ordered configurations concentrate in well-separated compact regions, while high-temperature disordered states occupy a large diffuse cluster in the upper portion of the latent space, consistent with the strong predictive performance of the model on $T^{(0)}$. The DMI strength $\tilde{K}_{DM}$ also produces a structured gradient in the latent space, with high-$\tilde{K}_{DM}$ helical configurations forming a distinct elongated region separated from the low-$\tilde{K}_{DM}$ disordered states, confirming that Xception has successfully encoded the chiral winding imposed by the DMI as a geometrically meaningful direction in its latent representation. The bulk anisotropy $\tilde{K}_{an1}$ and the external field $\tilde{H}_{ex}$ likewise display coherent, structured gradients across the embedding, mirroring their high regression accuracy and confirming that Xception encodes them as well-defined directions in its latent geometry — a direct consequence of the out-of-plane field configurations and the extended anisotropy range, which render these parameters observable in the s_z projection. In contrast, the longer-range exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ produce nearly uniform colorings across the entire embedding, indicating that the model's internal representations carry little discriminative information about these parameters — a result that is fully consistent with their

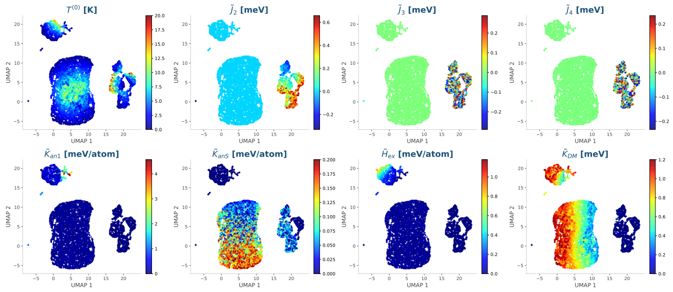

Figure 10: UMAP-based 2D projection of the Xception latent space on the held-out test set, colored by each of the eight Hamiltonian parameters $\boldsymbol{\theta} = [T^{(0)}, \tilde{J}_2, \tilde{J}_3, \tilde{J}_4, \tilde{K}_{an1}, \tilde{K}_{anS}, \tilde{H}_{ex}, \tilde{K}_{DM}]$. Each point represents a magnetization image $\mathbf{X}_n$ projected from the high-dimensional feature space of the Xception backbone into two dimensions via UMAP.

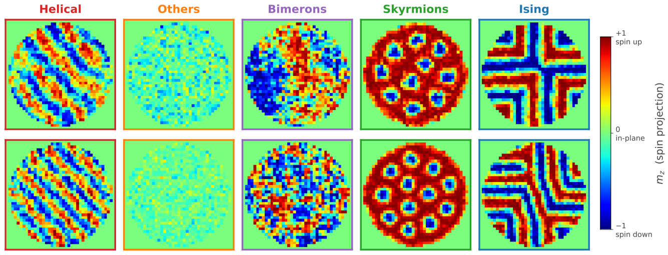

Figure 11: Representative magnetization images $\mathbf{X}_n$ for the five magnetic phase categories identified by clustering the Xception latent space. Each column is one phase (two samples nearest the cluster centroid in the UMAP embedding); color encodes the out-of-plane component m_z (+1 spin up, −1 spin down). The phases span ordered helical stripes, thermally disordered *others*, mixed-chirality bimerons, dense skyrmion lattices, and sharp Ising domain walls.

low R^2 values and their energetic subordination to the dominant first-shell exchange $\tilde{J}_1 = 1.0$ meV. The surface anisotropy $\tilde{K}_{anS}$ occupies an intermediate case, showing a partially mixed coloring consistent with its moderate identifiability and its entanglement with the bulk anisotropy signature.

Beyond the per-parameter gradients, the same embedding reveals that Xception has organized the magnetization textures into geometrically distinct and physically interpretable regions, without any explicit supervision of magnetic phase labels. The systematic variation of $T^{(0)}$ separates ordered, low-temperature configurations—which concentrate in compact, well-separated regions—from the thermally disordered states that occupy a large, diffuse cluster at high $T^{(0)}$, where thermal fluctuations dominate over the competing energetic contributions of $\tilde{J}_1$ and $\tilde{K}_{DM}$. Superimposed on this temperature axis, the $\tilde{K}_{DM}$ gradient further resolves the ordered region into the chiral helical, dense skyrmionic, and collinear Ising-like regimes, while the mixed-chirality configurations form their own transitional zone. This spontaneous geometric separation motivates a structured clustering analysis, performed directly on the two-dimensional UMAP embedding of the test set feature vectors (min-max normalized to [0, 1] along each axis), which we use next to identify and characterize the dominant magnetic phases.

To systematically identify and characterize the dominant magnetic phases encoded in the latent space, a density-based clustering analysis was performed using the HDBSCAN algorithm on the UMAP embedding, followed by a KMeans subdivision of anomalously large clusters. Based on visual inspection of representative samples and the mean values of the Hamiltonian parameters $\boldsymbol{\theta}$ within each group, the resulting clusters were consolidated into five physically meaningful magnetic phase categories.

The *helical phase* contains the most spatially coherent and regularly ordered configurations in the dataset, displaying well-defined stripe and labyrinthine patterns with consistent chiral winding direction and spatial periodicity across the nanodot, stabilized at low annealing temperature by a moderate DMI ($\langle\tilde{K}_{DM}\rangle \approx 0.75$ meV). The *skyrmion phase* groups configurations in which the strong DMI ($\langle\tilde{K}_{DM}\rangle \approx 0.88$ meV) stabilizes dense lattices of isolated, radially symmetric topological solitons, clearly visible as arrays of circular cores of reversed s_z embedded in a uniform background. The *Ising phase* comprises the low-temperature ($\langle T^{(0)}\rangle \approx 1.86$ K), high-anisotropy configurations characterized by sharp, discrete domain walls separating collinear up/down regions, reflecting the deliberately extended anisotropy range that drives the system toward an Ising-like regime. The *bimeron phase* gathers mixed-chirality textures in which in-plane and out-of-plane components coexist as paired meron-antimeron structures, forming irregular large-scale domains without the long-range periodicity of the helical phase. Finally, the *others* category constitutes the most heterogeneous group, gathering thermally disordered states in which thermal fluctuations have disrupted long-range spin order, transitional configurations located at the boundaries between competing phases, and topologically mixed textures whose spatial structure is ambiguous. The common characteristic of this group is the absence of a distinctive and reproducible magnetization pattern, making them the most structurally ill-conditioned samples in the dataset.

Figure 11 presents five representative magnetization images per phase, selected as the samples closest to the centroid of each group in the UMAP embedding, along with the mean Hamiltonian parameters $\langle T^{(0)}\rangle$, $\langle\tilde{J}_2\rangle$, and $\langle\tilde{K}_{DM}\rangle$ characterizing each phase.

The per-phase regression performance of the best-performing model, Xception, is summarized in Figures 12 and 13. The results reveal a pronounced heterogeneity in predictive accuracy across magnetic phases that is physically consistent with the identifiability analysis presented earlier in this section. The *skyrmion* and *Ising* phases achieve the highest R^2 values (typically 0.94–0.99) and the lowest MAE across essentially all identifiable parameters, reflecting the fact that these highly ordered configurations—dense soliton lattices and sharp, discrete domain walls, respectively—exhibit the most discriminative and spatially coherent signatures in the s_z projection. Within these phases, the well-defined periodicity and topological structure of the magnetization texture provide unambiguous observational constraints that allow the model to recover the underlying Hamiltonian parameters with high fidelity. The *helical* phase follows closely, with excellent recovery of $T^{(0)}$ and $\tilde{K}_{DM}$ ($R^2 \approx 0.99$) and good performance for $\tilde{J}_2$ and $\tilde{H}_{ex}$, although its anisotropy constants are estimated less accurately owing to the weaker anisotropic imprint in the chiral stripe regime.

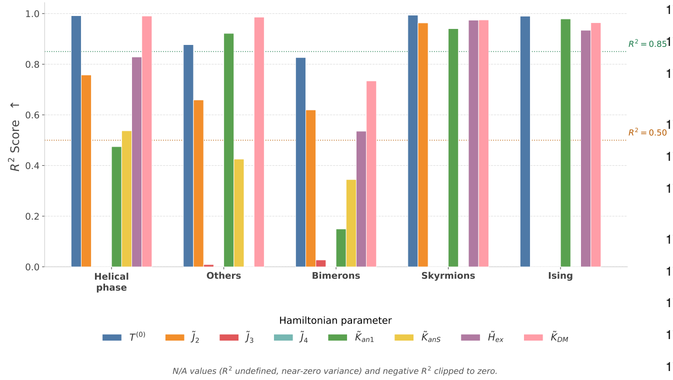

Figure 12: Per-phase R^2 score of Xception across the eight Hamiltonian parameters $\boldsymbol{\theta}$ for the five magnetic phases. Each bar group is one phase, each bar one parameter; dashed lines mark the $R^2 = 0.85$ (high) and $R^2 = 0.50$ (partial) predictability thresholds, and negative or near-zero-variance cases (N/A) are clipped to zero.

In stark contrast, the *others* and *bimeron* categories concentrate the largest absolute errors across virtually all Hamiltonian parameters, as evidenced by the dominant MAE values in Figure 13. This result directly confirms the physical interpretation advanced earlier: the thermally disordered configurations grouped in the *others* category, and the mixed-chirality meron-antimeron textures of the *bimeron* phase, relax into structurally weakly-discriminative magnetization patterns, so that widely different parameter combinations produce statistically similar textures and inflate the inverse-mapping error within these regimes. The elevated MAE for $T^{(0)}$ in the *others* phase is particularly revealing, as it quantitatively isolates the contribution of the high-temperature paramagnetic regime to the global prediction error observed in the aggregated analysis; likewise, the *bimeron* phase yields the lowest R^2 for the anisotropy and field parameters, reflecting the intrinsic spatial ambiguity of its coexisting in-plane and out-of-plane components.

Across all five phases, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ retain near-zero R^2, confirming at the per-phase level that their unidentifiability is a fundamental property of the s_z observation rather than an artifact of any particular structural regime.

5.3. RAMs-based Interpretability Results

To spatially trace the predictive behavior of the Xception architecture across the identified magnetic phase categories, RAMs were computed following the CNN-based formulation introduced in Section 3.4.1. For each of the five magnetic phases a single representative magnetization image $\mathbf{X}_n$ was selected as the sample closest to the centroid of the corresponding cluster group in the UMAP latent space, maximizing the concentration of gradient activations within the physical extent of the nanodot disk. For the spatial overlays, RAMs were computed for three representative Hamiltonian parameters ($T^{(0)}$, $\tilde{J}_2$ and $\tilde{K}_{DM}$) across eight strategically selected convolutional layers of the Xception backbone, spanning the full depth of the network:

– `block1_conv2`: entry-level features at spatial resolution 112×112, capturing low-level pixel-scale magnetization gradients.

– `block3_sepconv2` and `block4_sepconv2`: intermediate entry-flow layers at 56×56 and 28×28, encoding domain boundary frequencies and local spin-winding patterns.

– `block6_sepconv2` through `block12_sepconv2`: middle-flow layers at 14×14, constituting the deepest repeated processing stage of Xception and encoding progressively more abstract topological representations of the magnetic texture.

– `block13_sepconv2`: exit-flow transition layer prior to global pooling, encoding high-level semantic representations of the global magnetic phase.

Selection of the precision factor γ.. The precision factor γ in the RAM objective $O_p = \exp\left(-\gamma\left(\theta_p - \hat{\theta}_p\right)^2\right)$ governs the selectivity of the gradient signal: small values render the objective nearly constant regardless of prediction quality, producing spatially uninformative maps, while excessively large values collapse the gradient to zero for parameters with non-negligible prediction error, yielding empty activation maps. To select γ in a principled and data-driven manner, a sensitivity analysis was conducted by evaluating the objective O_p at the empirical mean prediction error $\bar{\varepsilon}_p = \langle|\theta_p - \hat{\theta}_p|\rangle$ of each Hamiltonian parameter on the test set, for values spanning $\gamma \in \{0.1, 0.3, 0.5, 1, 2, 5, 10, 20, 50, 100\}$.

Figure 14 illustrates the resulting sensitivity curves. The left panel shows the shape of the Gaussian objective as a function of the prediction error ε for each candidate γ: values below $\gamma = 1$ produce flat curves that activate almost uniformly regardless of prediction quality, while values above $\gamma = 20$ generate overly steep curves that suppress the gradient for errors as small as $\varepsilon \approx 0.2$; the selected $\gamma = 10$ provides an intermediate, well-graded response. The right panel evaluates O_p at the empirical mean error of each parameter as a function of γ: because the identifiable parameters ($T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$) are recovered with small mean errors, their objective remains close to unity across the entire γ range and never approaches the gradient collapse threshold $O_p < 0.05$. Only the poorly identified higher-order exchange constant $\tilde{J}_3$, whose mean error is comparatively large, shows a noticeable decline at high γ (down to $O_p \approx 0.8$ at $\gamma = 100$), yet even it stays well above the collapse regime. The value $\gamma = 10$ was therefore selected as the operating point: it is selective enough to yield spatially discriminative maps (steep left-panel response) while keeping a well-conditioned gradient signal for all Hamiltonian parameters.

To ensure that activation magnitudes remain physically interpretable and spatially confined to the nanodot geometry, each RAM was masked by a circular binary mask aligned with the disk boundary prior to normalization. The globally normalized RAMs $\tilde{\mathcal{R}}^{(l,p)}_{\mathrm{RAM}}$ were obtained following Equation 18, and the spatial patterns were additionally visualized under local per-cell normalization to reveal the gradient attribution structure at each individual layer.

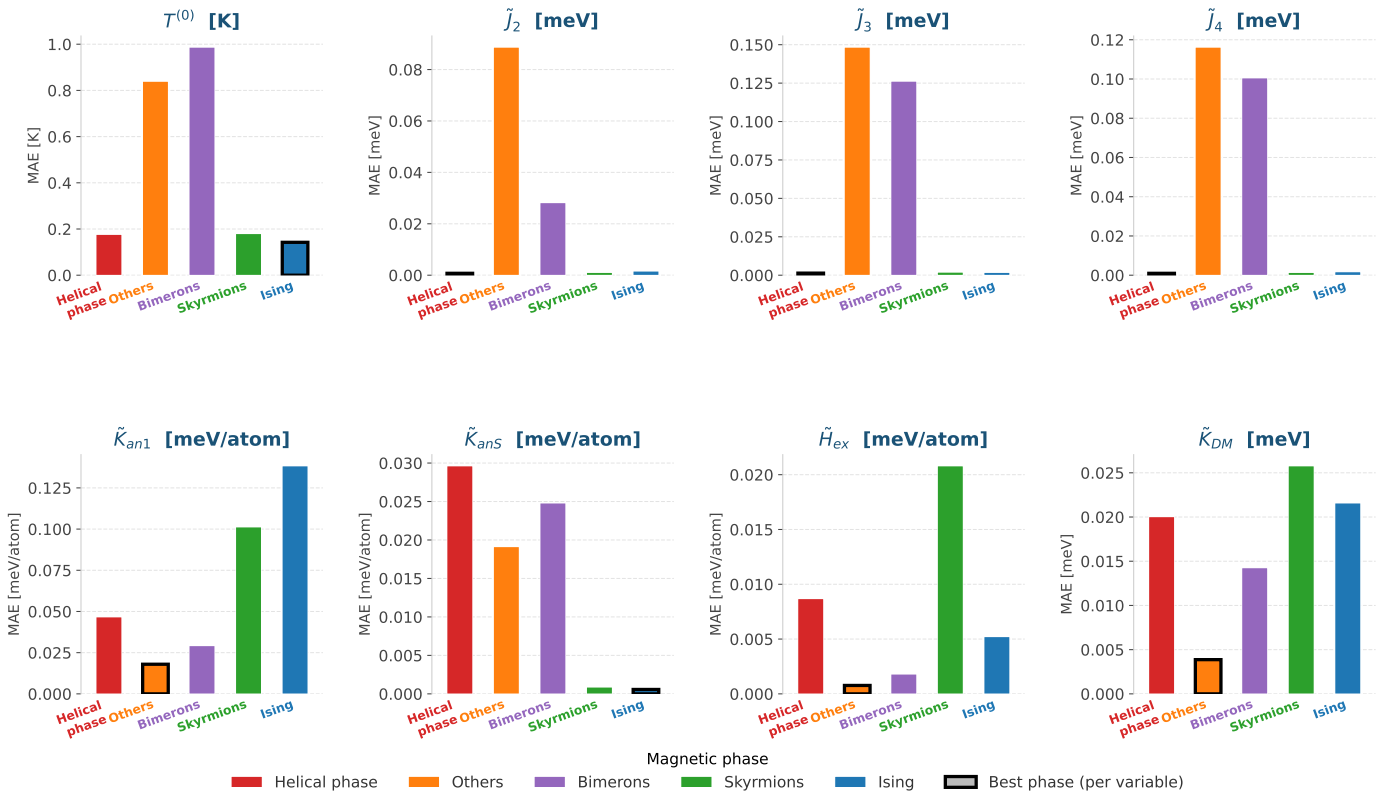

Figure 13: Per-phase Mean Absolute Error (MAE) of Xception across the eight Hamiltonian parameters $\boldsymbol{\theta}$ for the five magnetic phases, in each parameter's physical units. Each panel is one parameter, one bar per phase; black-bordered bars mark the phase with the lowest MAE. The *others* and *bimeron* categories concentrate the largest errors across most parameters.

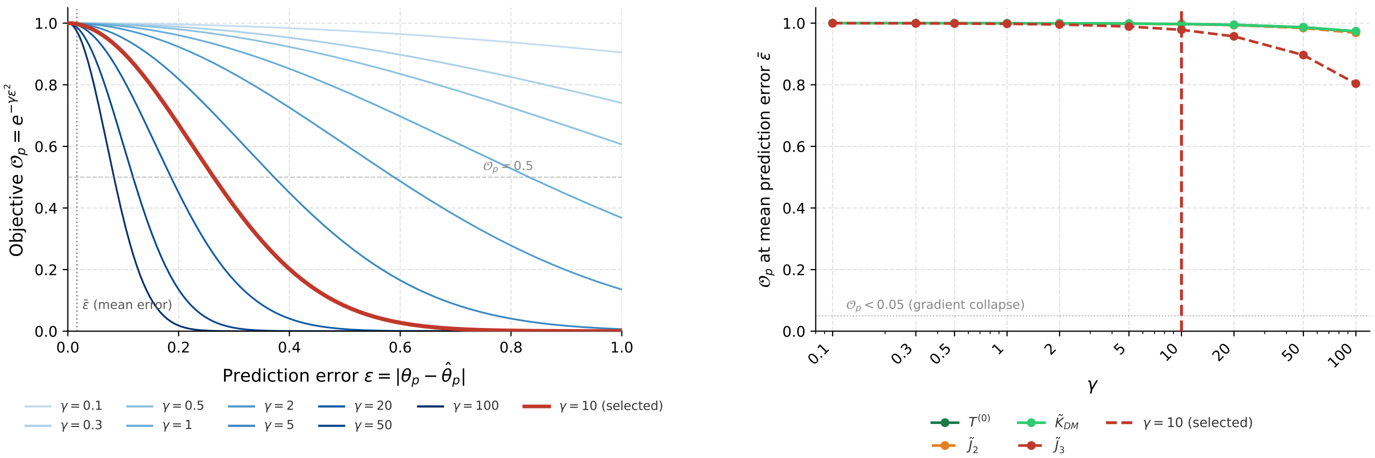

Figure 14: Sensitivity analysis of the RAM precision factor γ. Left: shape of the Gaussian objective $O_p = e^{-\gamma\varepsilon^2}$ versus the prediction error $\varepsilon = |\theta_p - \hat{\theta}_p|$ for $\gamma \in \{0.1, \ldots, 100\}$ (selected $\gamma = 10$ in red; vertical dotted line = empirical mean error). Right: O_p evaluated at each parameter's mean error as a function of γ; the horizontal dotted line marks the gradient-collapse threshold $O_p = 0.05$. The value $\gamma = 10$ is the selected operating point.

Figure 15 presents the resulting RAM overlays for the skyrmion phase, selected for visualization because its dense lattice of isolated, radially symmetric solitons produces the clearest and most physically interpretable gradient attribution patterns, providing the strongest validation of the hierarchical decoupling hypothesis. The spatial attribution patterns reveal a physically consistent progression across network depth. In the early entry-flow layers (`block1_conv2`, `block3_sepconv2`), the activations for $\tilde{K}_{DM}$ are spatially dense and sharply structured, coinciding with the individual skyrmion cores and their boundaries — precisely the chiral features that encode the DMI strength. As depth increases through the middle-flow layers (`block6` through `block12`), these activations progressively consolidate into coarser, spatially localized hotspots centered on the topologically significant skyrmion cores, while the $T^{(0)}$ attribution spreads over the inter-core background that reflects the global degree of thermal order. In the terminal exit-flow layer (`block13_sepconv2`), activations become concentrated in a small number of compact regions, indicating that the network's final semantic representations encode global topological stability rather than local texture periodicity. Consistently, the $\tilde{J}_2$ row remains nearly inactive across all layers, in agreement with its vanishing value in this phase. This hierarchical progression matches the theoretical prediction of Section 3.4.3: early layers resolve the short-range energetic competition between $\tilde{J}_1$ and $\tilde{K}_{DM}$, while deeper layers integrate these local features into macro-scale thermodynamic representations governed by $T^{(0)}$.

To quantify the relative importance of each network layer and each Hamiltonian parameter in driving the gradient response, Figure 16 presents the mean maximum RAM activation per layer and variable, averaged over $N = 300$ randomly sampled test images per magnetic phase. To enable meaningful cross-layer and cross-parameter comparisons, the raw activation values were standardized via a global z-score transform across all layers and variables within each phase, and subsequently mapped through a sigmoid function $\sigma\left(\frac{x-\mu}{\sigma_x}\right) \in (0, 1)$, where μ and σ_x denote the global mean and standard deviation of the raw activations for each variable. This normalization preserves the relative ordering of activation magnitudes while compressing outliers and centering the scale around $\sigma = 0.5$, such that values above this threshold indicate above-average gradient response for that parameter at that layer depth.

This statistical analysis confirms and extends the qualitative observations from the spatial maps. Across all five phases, the sigmoid-normalized activation profile exhibits a non-monotonic behavior across depth, with a consistently dominant peak at the terminal exit-flow layer `block13_sepconv2`, where the network aggregates the high-level semantic representations of the identifiable parameters ($T^{(0)}$, $\tilde{J}_2$, $\tilde{K}_{DM}$, $\tilde{K}_{an1}$, and $\tilde{H}_{ex}$). Secondary peaks in the middle-flow layers (`block6`–`block8`) reflect the intermediate stage at which the localized chiral and domain-wall features are assembled, and are most pronounced in the ordered skyrmion and Ising phases whose sharp textures provide the strongest spatial signal. By contrast, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ exhibit the lowest and least structured sigmoid activations across all layers and phases, corroborating their fundamental unidentifiability established in Section 5.2 and confirming that the Xception backbone has not learned spatially distinctive features for these parameters — a direct consequence of their energetic subordination to the dominant first- and second-shell exchange. The identifiable anisotropy and field parameters, in turn, attain their strongest activations in the phases where they leave a measurable imprint (the high-anisotropy Ising regime and the out-of-plane-field configurations), consistent with their successful recovery in the regression analysis.

The RAM analysis thus isolates a physically coherent set of features as the primary drivers of accurate regression: chiral domain-wall boundaries, stripe periodicity, and skyrmion cores. Conversely, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ never acquire spatially concentrated activations, confirming that their limited identifiability is not a consequence of insufficient model capacity but of the fundamental physical constraints of the inverse problem—their energetic subordination to the dominant first- and second-shell exchange, which leaves no distinguishable imprint on the s_z projection. For completeness, the RAM formulation was also derived for the Vision Transformer (Section 3.4.2); since Xception is the best-performing model and provides the clearest spatial attributions, the remainder of the interpretability analysis focuses on it.

5.4. Interpretability Benchmark: RAMs versus Perturbation-Based Attribution

Having established that RAMs isolate physically meaningful spatial features, we now benchmark them quantitatively against the perturbation-based attribution baselines introduced in Section 3.4.4—LIME, SHAP, and SF-SHAP—using the deletion protocol of Equations 22–23. The comparison targets both fidelity (how faithfully each map localizes the pixels that drive the prediction) and computational cost.

The perturbation baselines operate on superpixels rather than individual pixels, so the spatial granularity of the segmentation directly conditions their behavior. To fix this granularity on a physical basis, the number of SLIC segments was chosen so that the resulting superpixel equivalent diameter matches the characteristic magnetic domain width of the dataset (≈ 4.9 px). As shown in Figure 17, this criterion is satisfied at $N = 100$ segments, which we adopt for all perturbation baselines, ensuring that each interpretable region corresponds to one coherent magnetic domain rather than to an arbitrary pixel cluster.

Figure 18 reports the deletion curves for the three most representative parameters, and Figure 19 resolves the corresponding deletion AUC per magnetic phase. Two physically consistent trends emerge. First, for the spatially localized parameters—most notably the DMI strength $\tilde{K}_{DM}$—all attribution methods, including RAMs, degrade the prediction far more rapidly than the random-relevance baseline, and they do so most strongly in the ordered skyrmion and Ising phases where the chiral features are sharpest. This confirms that RAMs genuinely localize the pixels that carry the DMI signal, with a deletion AUC of the same order as the perturbation methods. Second, for the global thermodynamic parameter $T^{(0)}$ the situation is qualitatively reversed: no attribution method separates from—and the random baseline in fact matches or exceeds—the localized methods. This behavior is physically expected rather than a shortcoming of the attributions: the annealing temperature is not associated with any specific sector of the magnetic domain but is an *ambient* property encoded in the overall degree of order of the texture. Consequently, general (random) deletions distributed across the whole nanodot perturb the global order more effectively than deletions concentrated on a few "most relevant" regions, so that attempting to localize $T^{(0)}$ offers no advantage over uniform removal. The deletion metric thus correctly reflects that $T^{(0)}$ is a delocalized, environment-like parameter, in contrast to the spatially bound signatures of $\tilde{J}_2$ and $\tilde{K}_{DM}$.

The decisive advantage of RAMs lies in their computational cost. Figure 20 contrasts the cost of producing a sin-

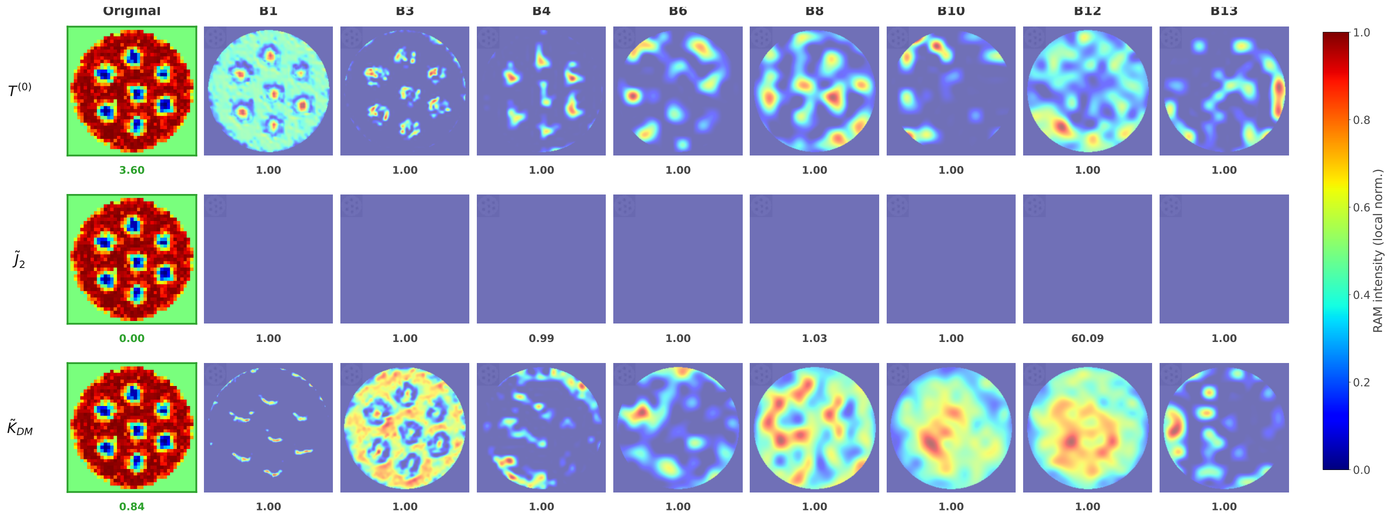

Figure 15: RAMs $\tilde{\mathcal{R}}^{(l,p)}_{\text{RAM}}$ for the Xception architecture in the skyrmion phase. Rows are three Hamiltonian parameters $\{T^{(0)}, \tilde{J}_2, \tilde{K}_{DM}\}$; columns are network layers from `block1` (entry, 112×112) to `block13` (exit, 14×14), with the original s_z image in the leftmost column. Each panel overlays the RAM (jet colormap) on the magnetization, locally normalized per layer. The score below each cell is the fraction of activation within the nanodot disk; the value below each original image is the ground-truth parameter.

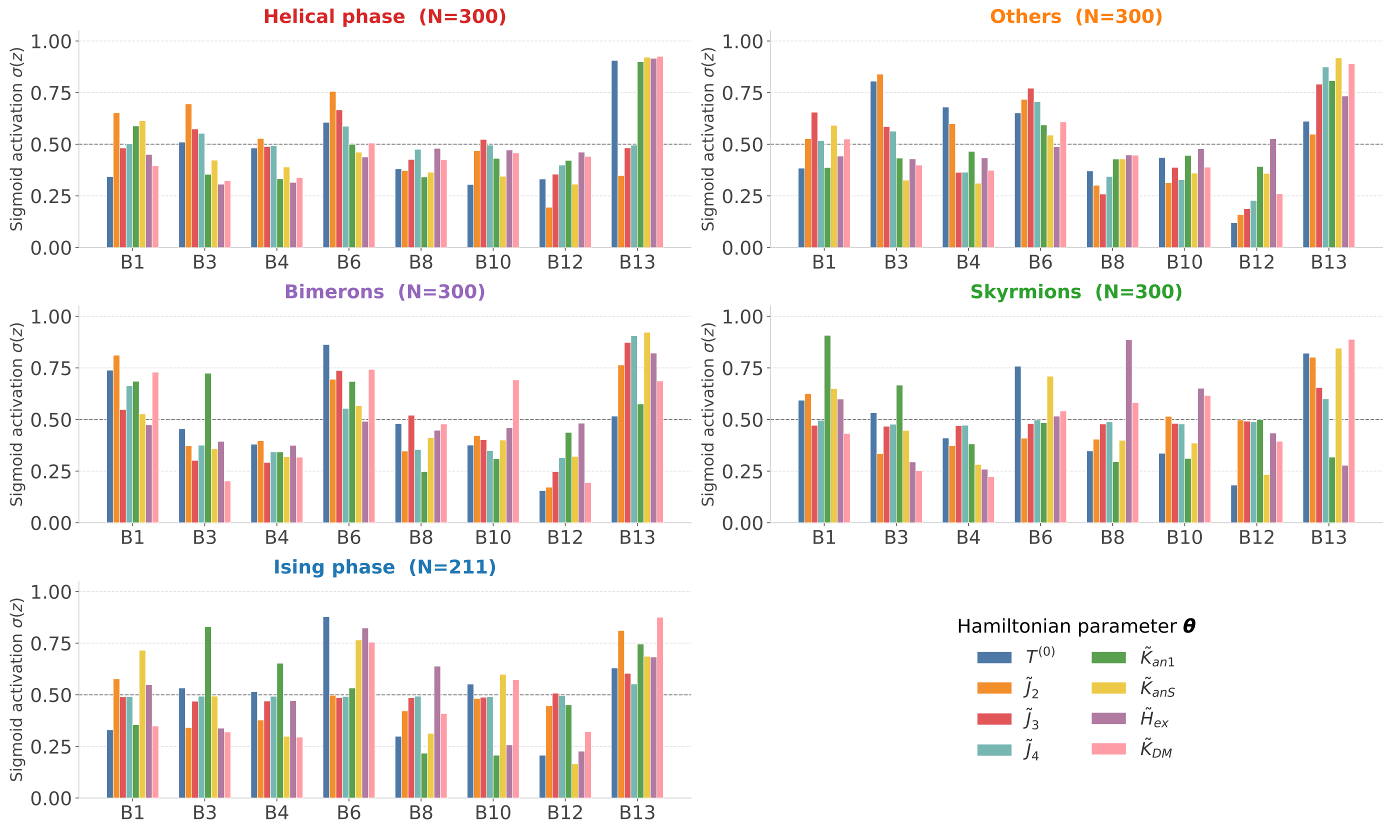

Figure 16: Mean maximum RAM activation per network layer and Hamiltonian parameter $\boldsymbol{\theta}$, averaged over $N = 300$ test samples per phase ($N = 211$ for Ising; Xception CNN), measured within the nanodot disk (masked RAM). Activations use a per-variable sigmoid normalization $\sigma\left(\frac{x-\mu}{\sigma_x}\right)$, where values above $\sigma = 0.5$ indicate above-average response; colors follow the parameter nomenclature of Table 2. The higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ show the lowest activation across all layers and phases, confirming their fundamental identifiability limit.

gle relevance map. Being gradient-based, a RAM requires exactly one backward pass—i.e., a single model evaluation—whereas the perturbation baselines must query the model over many masked coalitions: of order 10^3 evaluations for LIME and SHAP and $2^7 = 128$ for SF-SHAP. This translates into a measured wall-clock cost per map that is one to two orders of magnitude lower for RAMs than for any perturbation method. RAMs therefore deliver a faithfulness competitive with established attribution techniques while remaining tractable for the high-throughput, per-sample interpretability

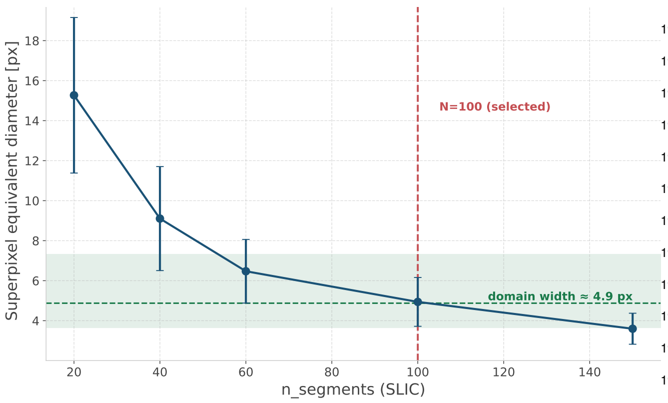

Figure 17: Physical criterion for the SLIC segmentation used by the perturbation-based baselines. The superpixel equivalent diameter is plotted against the number of SLIC segments; the selected value $N = 100$ (dashed red) yields superpixels whose size matches the characteristic magnetic domain width ($\approx$ 4.9 px, dashed green), so that each interpretable region corresponds to a single magnetic domain.

that materials-characterization pipelines require—an efficiency that the perturbation-based methods, by construction, cannot match.

5.5. Limitations

Despite the promising results demonstrated across the regression, clustering, and interpretability analyses, the proposed framework presents several inherent limitations that must be explicitly acknowledged to contextualize its applicability and guide future research directions.

The most fundamental limitation of the framework is the intrinsic partial observability of the extended Heisenberg Hamiltonian from two-dimensional magnetization images. As established in Section 5.2 and confirmed by the RAM-based interpretability analysis of Section 5.3, the projection of the three-dimensional spin configuration onto the out-of-plane component s_z of the central layer of the nanodot constitutes a dimensionality reduction that irreversibly discards physical information. While the dataset enrichment adopted in this work—the out-of-plane field configurations and the extended anisotropy range—renders six of the eight parameters recoverable, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ remain fundamentally unidentifiable: their energetic contribution is subordinated to the dominant first- and second-shell exchange, so they leave no distinguishable signature in the s_z texture, regardless of model capacity or architecture. A related, milder limitation affects the surface anisotropy $\tilde{K}_{anS}$, which is recovered only partially because the input images are extracted exclusively from the central discrete layer ($z = \lfloor \check{L}/2 \rfloor$) of the three-dimensional nanodot lattice, discarding the depth-resolved spin structure that encodes the surface-to-bulk anisotropy ratio $\tilde{K}_{anS}/\tilde{K}_{an1}$. Recovering these remaining parameters with high fidelity would require either access to the full three-dimensional spin vector field $\mathbf{S}_i = (s_x, s_y, s_z)$ at each lattice site, or volumetric magnetization inputs obtained through multi-layer or tomographic magnetic imaging techniques. This is not a limitation of the deep learning models themselves but of the observational modality assumed by the framework, which mimics the two-dimensional projection inherent to most experimental magnetic imaging techniques such as magnetic force microscopy or Lorentz transmission electron microscopy.

The dataset utilized in this study consists entirely of simulated magnetization images generated through the atomistic Monte Carlo framework described in Section 3.2. While the simulation protocol incorporates physically realistic finite-size effects, cylindrical boundary conditions, and thermodynamic relaxation via simulated annealing, it does not account for several experimental phenomena that are routinely present in real magnetic nanodot imaging. These include thermal noise and instrument-specific point spread functions inherent to the imaging apparatus, crystallographic defects, vacancies, and grain boundaries that break the assumed periodic lattice symmetry, surface reconstruction effects that modify the effective exchange and DMI parameters at the physical boundary of the nanodot, and substrate-induced strain fields that can renormalize the bulk Hamiltonian parameters. As a consequence, a model trained exclusively on simulated data may exhibit significant performance degradation when applied to experimentally acquired magnetization images — a domain shift problem that has not been quantified in the present work. Bridging this simulation-to-experiment gap would require either transfer learning with a small set of experimentally labeled samples or the incorporation of physics-informed data augmentation strategies that realistically replicate experimental noise sources.

A further limitation concerns the internal structure of the sampled parameter space. The dataset spans two mechanisms for stabilizing modulated textures—exchange frustration (driven by $\tilde{J}_2$) and the chiral DMI (driven by $\tilde{K}_{DM}$)—which were deliberately activated in a mutually exclusive fashion to isolate and compare their individual contributions, and were subsequently unified into a single training corpus. While this design cleanly separates the two mechanisms and demonstrates that both can independently generate analogous modulated states, it does not include configurations in which the two interactions coexist and compete simultaneously, as may occur in real materials. Extending the dataset to jointly sample $\tilde{J}_2$ and $\tilde{K}_{DM}$ would allow the model to learn their combined effect and is a natural direction for future work.

From a deep learning perspective, the RAM-based interpretability framework introduced in this work provides spatially resolved gradient attributions that are physically consistent and architecturally agnostic. However, RAMs share the fundamental limitations of all gradient-based saliency methods: the attribution maps reflect the local sensitivity of the objective function O_p to spatial activations at the point of evaluation, but do not provide a globally faithful explanation of the model's decision function across the full parameter space. In particular, gradient saturation in deeply nonlinear networks can produce spatially diffuse or spurious attributions for parameters with low predictive signal — precisely the degenerate parameters identified in this study. Furthermore, the current RAM formulation computes a single attribution map per image per parameter, which does not capture the stochastic uncertainty in the attribution due to the non-deterministic elements of the forward pass. Future extensions could incorporate ensemble-based or Bayesian attri-

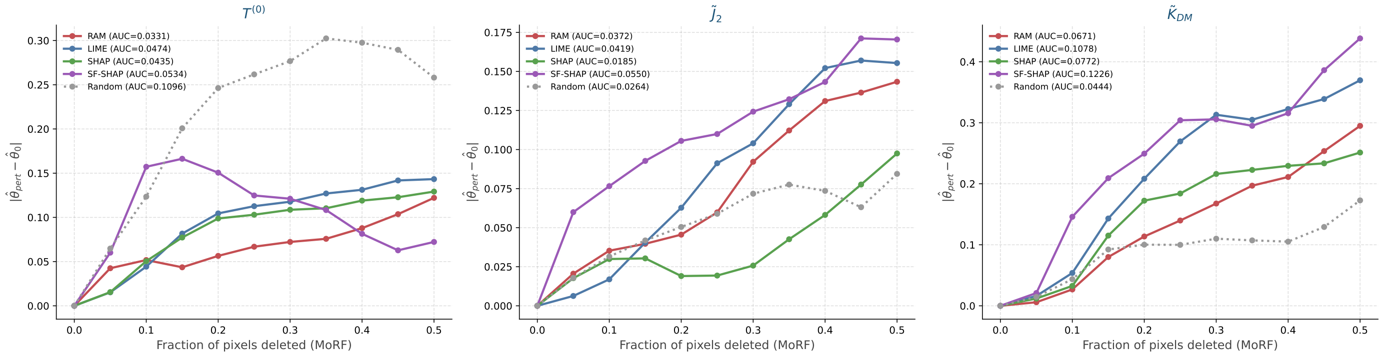

Figure 18: Deletion curves (Most Relevant First) for RAM, LIME, SHAP, and SF-SHAP, with a random-relevance lower bound, for the three representative parameters $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$. The vertical axis is the induced deviation $|\hat{\theta}_{\rm pert} - \hat{\theta}_0|$ as the most relevant pixels are progressively replaced by the baseline; the area under each curve (AUC, in the legend) quantifies faithfulness.

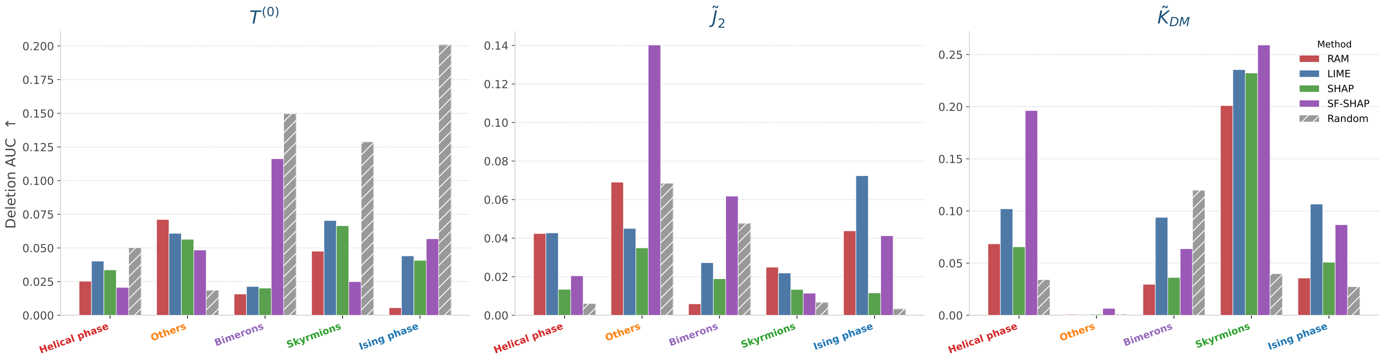

Figure 19: Deletion AUC resolved by magnetic phase for RAM, LIME, SHAP, and SF-SHAP (with the random baseline), for $T^{(0)}$, $\tilde{J}_2$, and $\tilde{K}_{DM}$; a higher AUC indicates a more faithful attribution.

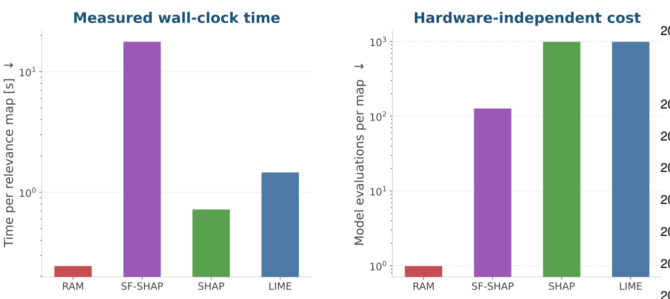

Figure 20: Computational cost of the interpretability methods. Left: measured wall-clock time per relevance map (log scale). Right: number of model evaluations per map (log scale). A RAM requires a single backward pass, against $\sim 10^3$ evaluations for LIME and SHAP and 128 for SF-SHAP.

bution methods to provide confidence intervals on the spatial interpretability maps, which would be particularly valuable for the physically ambiguous parameter–phase combinations identified in Section 5.3.

A more fundamental limitation is intrinsic to the observable itself: because the framework operates on the out-of-plane s_z projection, parameters whose signature lies in the in-plane components—or whose energetic contribution is subordinated to the dominant exchange, as for $\tilde{J}_3$ and $\tilde{J}_4$—cannot be recovered, a boundary that would require chirality-sensitive or fully three-dimensional observables to overcome.

6. Conclusions

This work introduced a regression-based explainable deep learning framework for the inverse estimation of Hamiltonian parameters from two-dimensional magnetization images of magnetic nanodots. The proposed methodology integrates atomistic Monte Carlo simulations under a physically rigorous extended Heisenberg Hamiltonian with eight free parameters, five state-of-the-art deep learning architectures, an unsupervised latent space analysis via UMAP and density-based clustering, and a novel spatial interpretability formulation RAMs that generalizes gradient-based activation mapping to the continuous regression domain for both convolutional and transformer-based architectures.

Among the five evaluated architectures, Xception achieved the highest overall regression accuracy, although all models—including the compact conventional CNN baseline trained from scratch—performed comparably, demonstrating that estimation accuracy is governed by the physical information available in the s_z projection rather than by model capacity or pretraining. Six of the eight Hamiltonian parameters were recovered with high fidelity: the annealing temperature $T^{(0)}$, the second-shell exchange $\tilde{J}_2$, the DMI strength $\tilde{K}_{DM}$, the bulk anisotropy $\tilde{K}_{an1}$, and the external Zeeman field $\tilde{H}_{ex}$, with the surface anisotropy $\tilde{K}_{anS}$ at an intermediate level. This broad identifiability was made possible by the deliberate enrichment of the dataset with out-of-plane field configurations—coupling the Zeeman term to the observable s_z component—and an extended anisotropy range. By contrast, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$ exhibited near-zero R^2 across all architectures, establishing that the fundamental limitation of the inverse problem is not architectural capacity but the intrinsic partial observability of these subordinate interactions from the two-dimensional projection.

Beyond the aggregate regression performance, the UMAP projection of the Xception latent representations revealed that the model autonomously organized the magnetization textures into geometrically distinct and physically interpretable regions of the embedding space, without any explicit supervision of magnetic phase labels. Density-based clustering identified five dominant magnetic phase categories—helical, skyrmion, Ising, bimeron, and a heterogeneous *others* group. The per-phase regression analysis confirmed that the model achieves its highest accuracy within the most ordered phases (skyrmion and Ising, $R^2 \gtrsim 0.94$ across the identifiable parameters), where the magnetization texture carries the most discriminative physical information, while the disordered and mixed-chirality configurations concentrate the largest errors.

The proposed RAMs formulation successfully extended gradient-based activation mapping to the continuous regression setting through a scale-invariant Gaussian error-decay objective $O_p = \exp(-\gamma_p(\theta_p - \hat{\theta}_p)^2)$, providing physically interpretable spatial attribution maps for each Hamiltonian parameter independently. For the Xception architecture, the layer-resolved RAM analysis revealed a physically consistent hierarchical decoupling across network depth: early entry-flow layers capture fine-grained domain periodicity and chiral winding at high spatial resolution, while deeper middle-flow and exit-flow layers consolidate these local features into coarser, topologically meaningful representations, with the DMI attributions localizing on the skyrmion cores that encode it. Benchmarked against the LIME, SHAP, and exact spatial-frequency Shapley baselines through a deletion protocol, RAMs attained a competitive spatial faithfulness—far above the random baseline for the localized parameters—while requiring a single backward pass per map, one to two orders of magnitude cheaper than the perturbation-based methods. This combination of physical faithfulness and computational efficiency makes RAMs practical for per-sample interpretability in high-throughput materials characterization.

Taken together, the results of this study demonstrate that regression-based deep learning, combined with physically motivated interpretability methods, constitutes a viable and physically meaningful approach to the inverse Hamiltonian estimation problem in magnetic nanostructures. The proposed RAMs framework is architecture-agnostic and directly applicable to any continuous regression task where spatial attribution of physical parameters is required, extending beyond magnetic nanodots to other physical systems characterized by spatially resolved imaging modalities.

Several directions merit investigation in future work. First, incorporating the full three-dimensional spin vector field $\mathbf{S}_i = (s_x, s_y, s_z)$ or multi-layer volumetric magnetization images as input would substantially reduce the partial observability constraint and improve the identifiability of the parameters that remain elusive under the s_z-only observation—the surface anisotropy $\tilde{K}_{anS}$ and, in particular, the higher-order exchange constants $\tilde{J}_3$ and $\tilde{J}_4$. Second, domain adaptation strategies, including physics-informed data augmentation and few-shot transfer learning from experimentally labeled samples, are necessary to bridge the simulation-to-experiment gap and enable direct application to real magnetic imaging data. Third, extending the RAM formulation to incorporate uncertainty quantification through ensemble-based or Bayesian attribution methods would provide confidence intervals on the spatial interpretability maps, which are particularly valuable for physically ambiguous parameter–phase combinations near topological phase boundaries. Finally, the integration of the proposed framework with active learning strategies, where the model identifies the most informative parameter combinations to simulate next, could dramatically accelerate the construction of training datasets for high-dimensional Hamiltonian spaces, enabling the extension of the inverse estimation approach to more complex magnetic systems with larger numbers of competing energetic contributions.

Acknowledgments

Under grants provided by the research project: "Aprendizaje automático aplicado a la predicción eficiente de propiedades magnéticas de nanopartículas como soporte a la generación de dispositivos electrónicos" Hermes 62642, funded by Universidad Nacional de Colombia. Also, A. Alvarez thanks to the project: "Estimación de la capacidad de generación de energía eléctrica del agua coproducida en campos de petróleo y gas en Colombia a partir de técnicas de aprendizaje automático informado por la física," Code 111908, funded by 951-2024 Convocatoria fortalecimiento del conocimiento científico y tecnológico de las fuentes - Minciencias.ñkm

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Gemini and Claude in order to enhance the writing style. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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